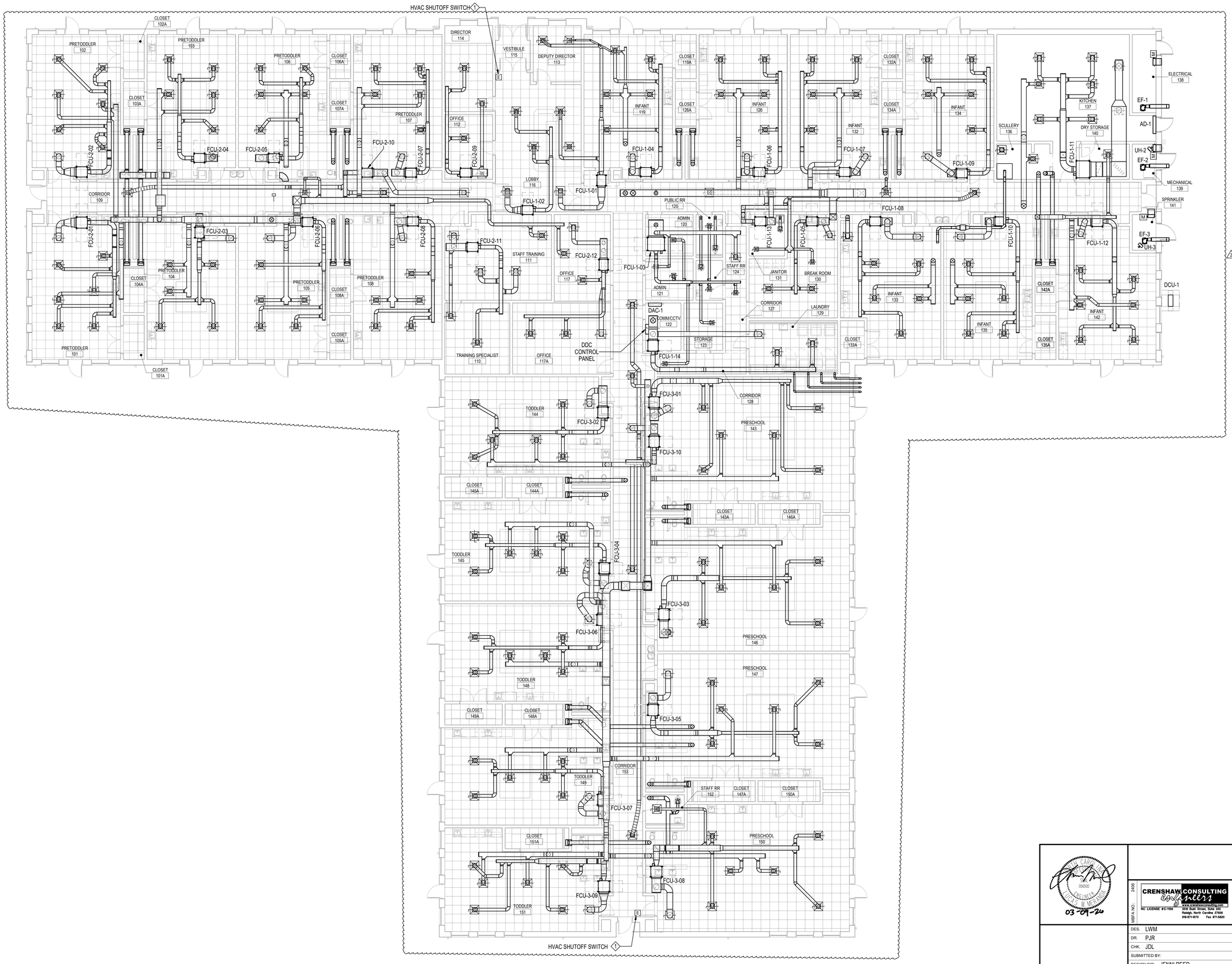
## REVISIONS

| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| 2 | PPI REVISION 2 | 03.09.26 | |
| | | | |
| | | | |




A1 FIRST FLOOR PLAN - OVERALL - MECHANICAL HVAC
1/8" = 1'-0"

### GENERAL NOTES:

1. SEE SHEET M-001 FOR GENERAL NOTES, LEGEND & ABBREVIATIONS
2. PLAN DRAWINGS ARE DIAGRAMMATIC IN NATURE AND DO NO ILLUSTRATE SPECIFIC DUCT TAKE-OFF CONFIGURATIONS OR TAPS.
3. FAN COIL UNITS AND ASSOCIATED DUCTWORK SHOWN IS LOCATED IN ATTIC AND SHOWN ON FLOOR PLANS FOR COORDINATION PURPOSES ONLY. SEE ATTIC AND SECTION VIEWS FOR ELEVATIONS AND LOCATIONS OF EQUIPMENT.

### # PLAN NOTES

1 PROVIDE WALL MOUNTED EMERGENCY HVAC SHUTDOWN SWITCH. SEE SHEET M-701 FOR MORE INFORMATION. SEE ENLARGED SHEETS FOR MORE INFORMATION.

GRAPHIC SCALE: 1/8"=1'-0"
8' 0 4' 8' 16'

MH101

03-09-26

MBFA NO.: 2406

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DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND

MARINE CORPS BASE

CAMP LEJEUNE, NORTH CAROLINA

DES. LWM
DR. PJR
CHK. JDL
SUBMITTED BY:
DESIGN DIR. JENNI REED
APPROVED: PWO OR OICC DATE
SATISFACTORY TO: DATE

MECHANICAL REPAIRS
BUILDING TT113

FIRST FLOOR PLAN - OVERALL - MECHANICAL HVAC

| SIZE | CODE IDENT. NO. | NAVFAC DRAWING NO. |
|---|---|---|
| E1 | 80091 | 60042066 |
| | | CONSTR. CONTR. NO. N40085-24-F-5940 |
| SCALE AS NOTED | SPEC. | SHEET 59 OF 119 |

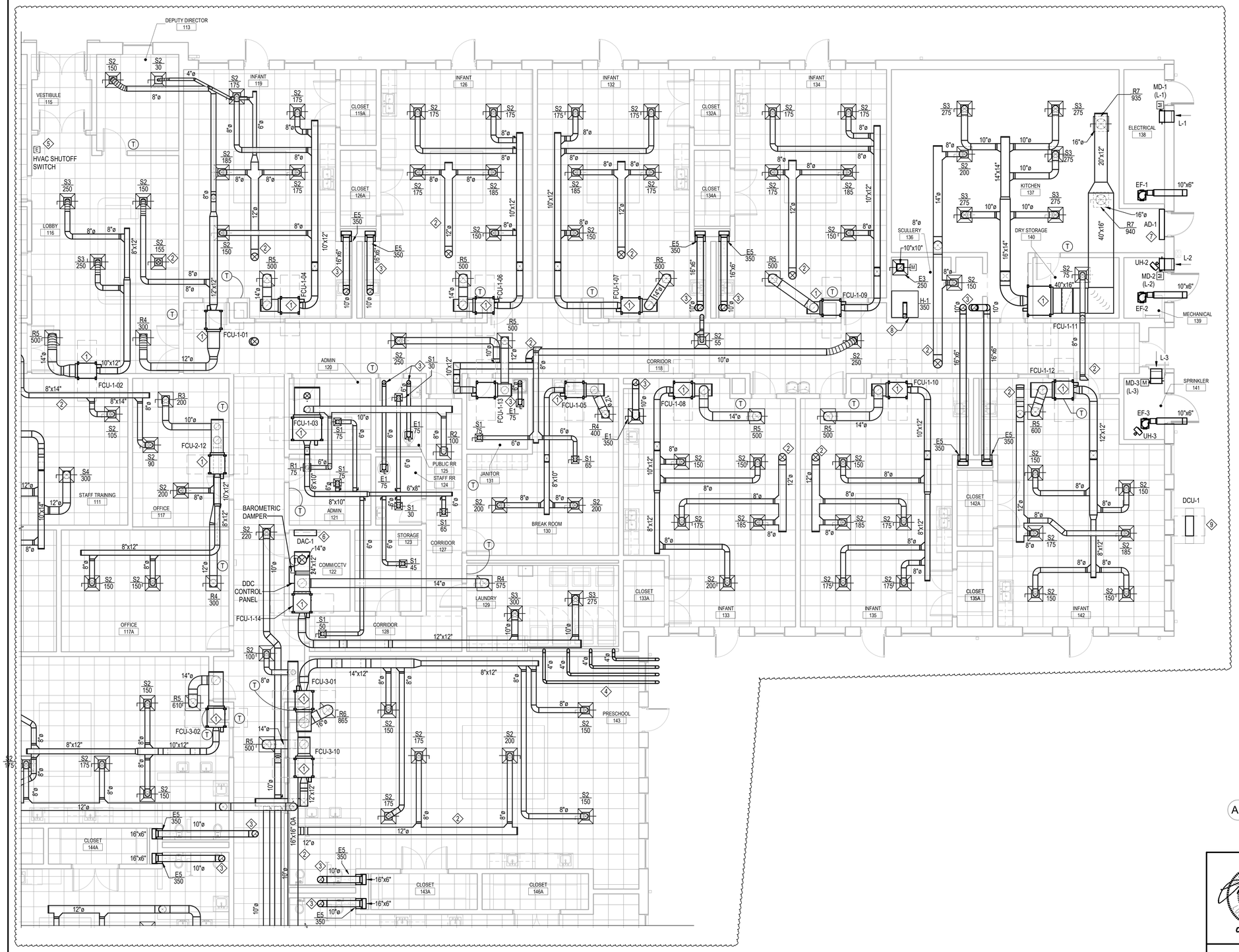
| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| 2 | PPI REVISION 2 | 03.09.26 | |

## GENERAL NOTES:

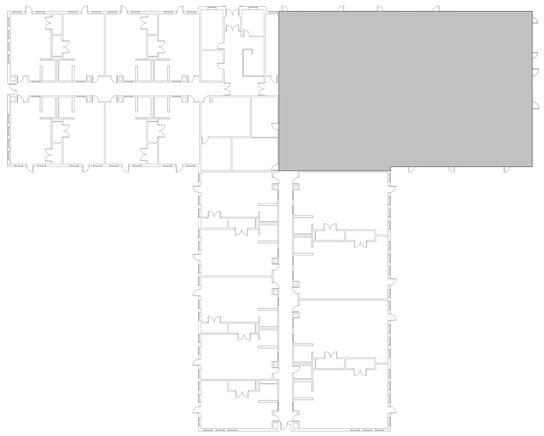
1. SEE SHEET M-001 FOR GENERAL NOTES, LEGEND & ABBREVIATIONS
2. PLAN DRAWINGS ARE DIAGRAMMATIC IN NATURE AND DO NO ILLUSTRATE SPECIFIC DUCT TAKE-OFF CONFIGURATIONS OR TAPS.
3. FAN COIL UNITS AND ASSOCIATED DUCTWORK SHOWN IS LOCATED IN ATTIC AND SHOWN ON FLOOR PLANS FOR COORDINATION PURPOSES ONLY. SEE ATTIC AND SECTION VIEWS FOR ELEVATIONS AND LOCATIONS OF EQUIPMENT.

## # PLAN NOTES

1. PROVIDE 4-PIPE FAN COIL UNIT ABOVE SECOND GYPSUM BOARD CEILING. PROVIDE RETURN DUCT PLENUM TO MATCH DUCT INLET CONNECTION. PROVIDE TRANSITION OR TAP AS SHOWN AND ROUTE SUPPLY AND RETURN DUCTWORK DOWN THROUGH GYP CEILING TO LAY-IN CEILING GRILLES. PROVIDE DUCT MOUNTED BALANCING DAMPER BETWEEN GYP CEILING AND GRILLE.
2. PROVIDE VENTILATION DUCTWORK DOWN FROM DOAS MAIN DUCTWORK IN ATTIC, TO LAY-IN CEILING MOUNTED GRILLES AS SHOWN. SEE ATTIC PLANS FOR CONTINUATION. (TYPICAL)
3. PROVIDE EXHAUST DUCTWORK DOWN FROM DOAS MAIN DUCTWORK IN ATTIC. ROUTE TO VERTICAL EXHAUST DUCTWORK IN CHASE DOWN TO SIDEWALL MOUNTED GRILLE AS SHOWN. (TYPICAL)
4. PROVIDE EXHAUST VENT DUCTWORK FROM DRYERS TO EXTERIOR WALL CAP(S) AS SHOWN. REUSE EXISTING PENETRATIONS.
5. PROVIDE WALL MOUNTED EMERGENCY HVAC SHUTDOWN SWITCH. PROVIDE SWITCH WITH A DIFFERENT COLOR THAN FIRE ALARM PULL STATIONS. SEE SHEET M-701 FOR MORE INFORMATION.
6. PROVIDE WALL MOUNTED DUCTLESS SPLIT AIR HANDLER ABOVE DOOR. PROVIDE MANUFACTURER'S RECOMMENDED CLEARANCES AND COORDINATE INSTALLATION WITH OTHER EQUIPMENT IN SPACE.
7. PROVIDE AIR DOOR ABOVE EXTERIOR DOOR. OPERATION MUST BE TIED TO DOOR SWITCH.
8. PROVIDE EXHAUST HOOD DUCTED UP TO EXHAUST FAN IN ATTIC. EXHAUST HOOD AND FAN OPERATION MUST BE TIED TO WALL MOUNTED SWITCH, AND ONLY BE ENERGIZED UPON PROOF OF CLOSURE OF MOTORIZED DAMPER IN DOAS EXHAUST DUCT SERVING SPACE.
9. PROVIDE CONDENSING UNIT MOUNTED ON EQUIPMENT SUPPORT PAD ON EXTERIOR OF BUILDING AS SHOWN. COORDINATE EXACT LOCATION WITH CIVIL AND ENSURE MANUFACTURER RECOMMENDED CLEARANCES ARE MAINTAINED.




A1 FIRST FLOOR PLAN - EAST WING - MECHANICAL HVAC
3/16" = 1'-0"



A5 KEY PLAN - EAST WING
NTS

GRAPHIC SCALE: 3/16"=1'-0"
6' 0 4' 10'

03-09-26

MH102

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MBFA NO.: 24/06

DEPARTMENT OF THE NAVY — NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND

MARINE CORPS BASE

CAMP LEJEUNE, NORTH CAROLINA

MECHANICAL REPAIRS
BUILDING TT113

FIRST FLOOR PLAN - EAST WING - MECHANICAL HVAC

DES. LWM
DR. PJR
CHK. JDL
SUBMITTED BY:
DESIGN DIR. JENNI REED
APPROVED: PWO OR OICC — DATE
SATISFACTORY TO: — DATE

| SIZE | CODE IDENT. NO. | NAVFAC DRAWING NO. |
|---|---|---|
| E1 | 80091 | 60042067 |
| SCALE AS NOTED | SPEC. | CONSTR. CONTR. NO. N40085-24-F-5940 |

SHEET 60 OF 119

## REVISIONS

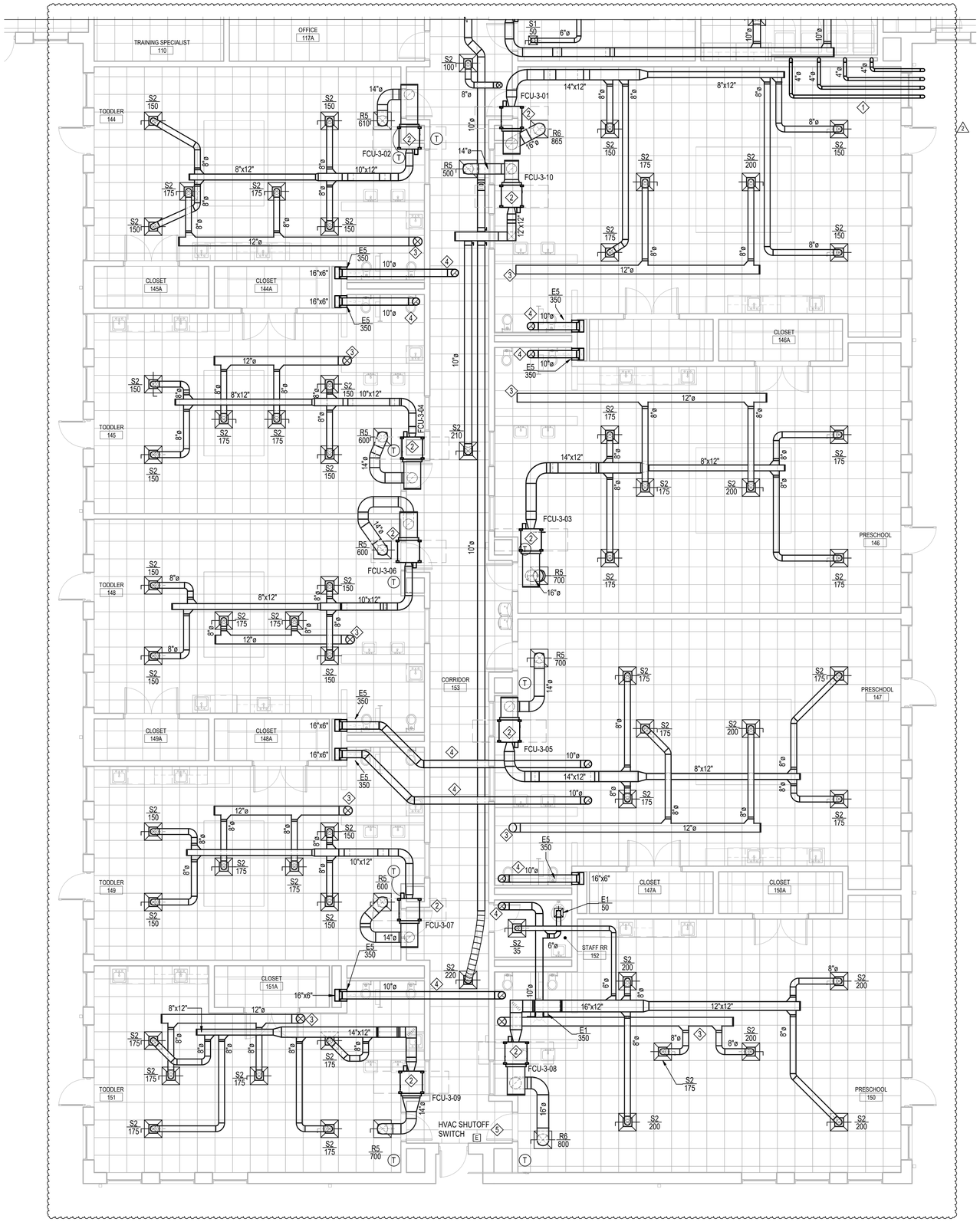
| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| 2 | PPI REVISION 2 | 03.09.26 | |
| | | | |
| | | | |

### GENERAL NOTES:

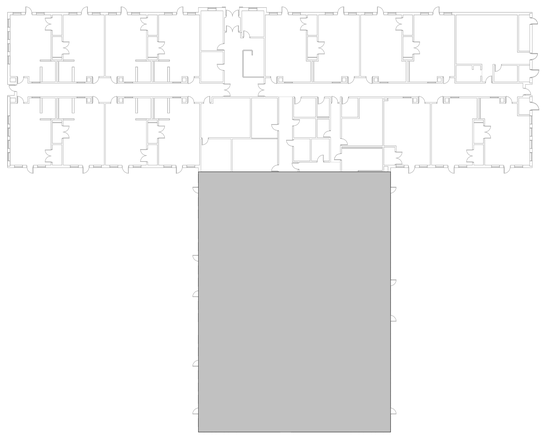
1. SEE SHEET M-001 FOR GENERAL NOTES, LEGEND & ABBREVIATIONS
2. PLAN DRAWINGS ARE DIAGRAMMATIC IN NATURE AND DO NO ILLUSTRATE SPECIFIC DUCT TAKE-OFF CONFIGURATIONS OR TAPS.
3. FAN COIL UNITS AND ASSOCIATED DUCTWORK SHOWN IS LOCATED IN ATTIC AND SHOWN ON FLOOR PLANS FOR COORDINATION PURPOSES ONLY. SEE ATTIC AND SECTION VIEWS FOR ELEVATIONS AND LOCATIONS OF EQUIPMENT.

### PLAN NOTES

1. PROVIDE EXHAUST VENT DUCTWORK FROM DRYERS TO EXTERIOR WALL CAP(S) AS SHOWN. REUSE EXISTING PENETRATIONS.
2. PROVIDE 4-PIPE FAN COIL UNIT ABOVE SECOND GYPSUM BOARD CEILING. PROVIDE RETURN DUCT PLENUM TO MATCH DUCT INLET CONNECTION. PROVIDE TRANSITION OR TAP AS SHOWN AND ROUTE SUPPLY AND RETURN DUCTWORK DOWN THROUGH GYP CEILING TO LAY-IN CEILING GRILLES. PROVIDE DUCT MOUNTED BALANCING DAMPER BETWEEN GYP CEILING AND GRILLE.
3. PROVIDE VENTILATION DUCTWORK DOWN FROM DOAS MAIN DUCTWORK IN ATTIC, TO LAY-IN CEILING MOUNTED GRILLES AS SHOWN. SEE ATTIC PLANS FOR CONTINUATION. (TYPICAL)
4. PROVIDE EXHAUST DUCTWORK DOWN FROM DOAS MAIN DUCTWORK IN ATTIC. ROUTE TO VERTICAL EXHAUST DUCTWORK IN CHASE DOWN TO SIDEWALL MOUNTED GRILLE AS SHOWN. (TYPICAL)
5. PROVIDE WALL MOUNTED EMERGENCY HVAC SHUTDOWN SWITCH. PROVIDE SWITCH WITH A DIFFERENT COLOR THAN FIRE ALARM PULL STATIONS. SEE SHEET M-701 FOR MORE INFORMATION.




A2 FIRST FLOOR PLAN - SOUTH WING - MECHANICAL HVAC
3/16" = 1'-0"



A5 KEY PLAN - SOUTH WING
NTS

GRAPHIC SCALE: 3/16"=1'-0"
6' 0 4' 10'

MH103

03-09-26

MBFA NO.: 2406
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DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND

MARINE CORPS BASE

CAMP LEJEUNE, NORTH CAROLINA

DES. LWM
DR. PJR
CHK. JDL
SUBMITTED BY:
DESIGN DIR. JENNI REED
APPROVED: PWO OR OICC DATE
SATISFACTORY TO: DATE

MECHANICAL REPAIRS
BUILDING TT113

FIRST FLOOR PLAN - SOUTH WING - MECHANICAL HVAC

| SIZE | CODE IDENT. NO. | NAVFAC DRAWING NO. |
|---|---|---|
| E1 | 80091 | 60042068 |
| | | CONSTR. CONTR. NO. N40085-24-F-5940 |

SCALE AS NOTED SPEC. SHEET 61 OF 119

## REVISIONS

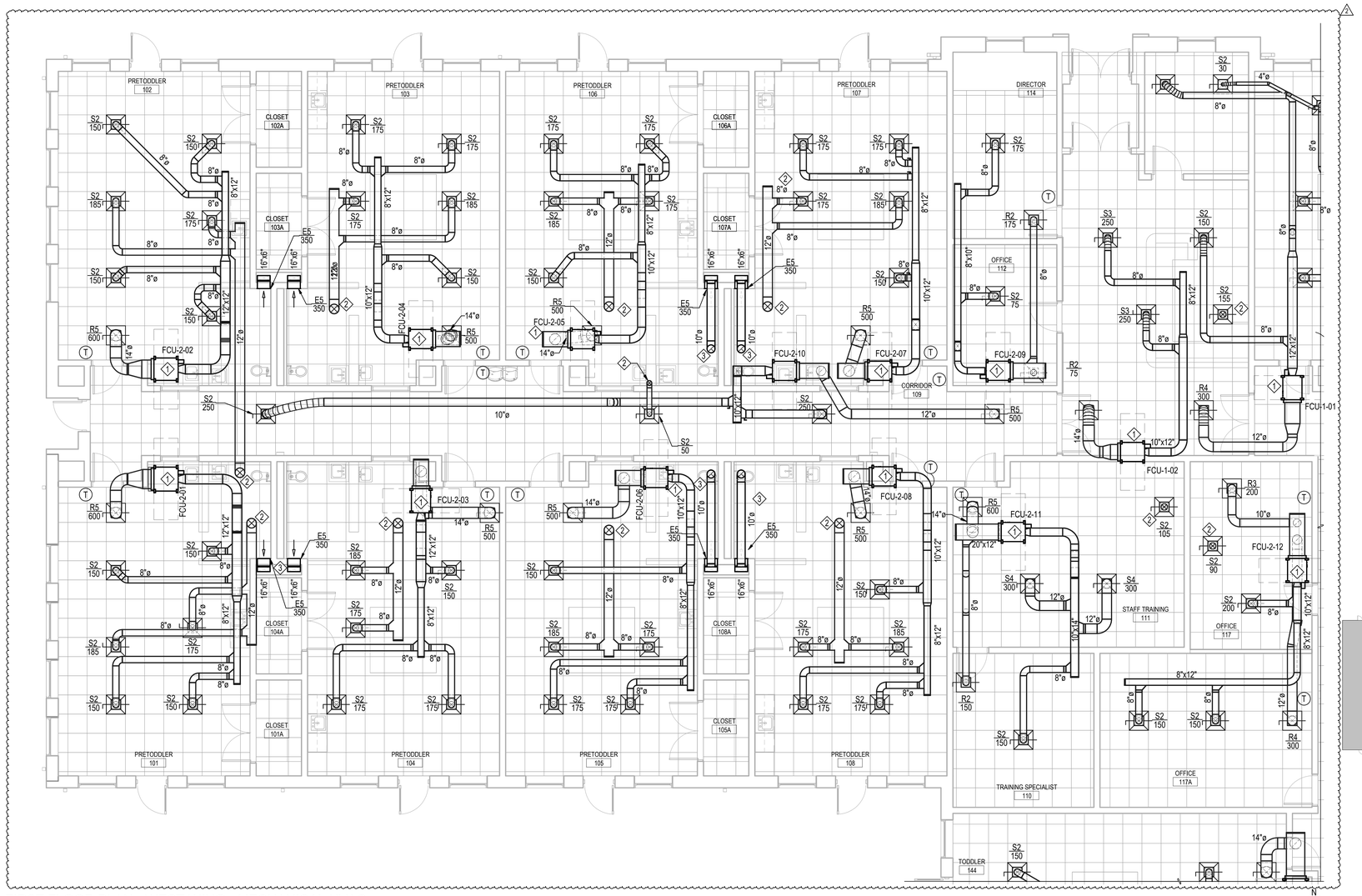
| SYM. | DESCRIPTION | DATE | APP. |
|---|---|---|---|
| 2 | PPI REVISION 2 | 03.09.26 | |
| | | | |
| | | | |

### GENERAL NOTES:

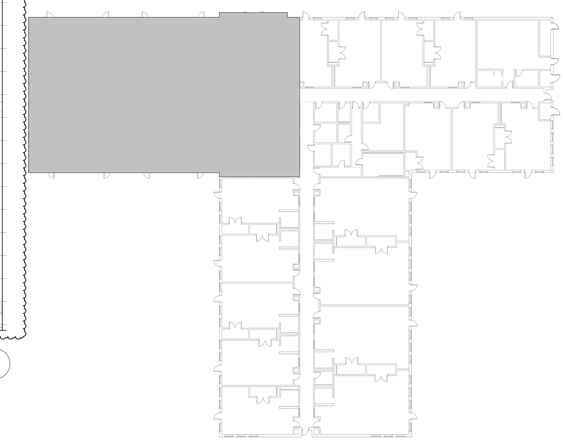
1. SEE SHEET M-001 FOR GENERAL NOTES, LEGEND & ABBREVIATIONS
2. PLAN DRAWINGS ARE DIAGRAMMATIC IN NATURE AND DO NO ILLUSTRATE SPECIFIC DUCT TAKE-OFF CONFIGURATIONS OR TAPS.
3. FAN COIL UNITS AND ASSOCIATED DUCTWORK SHOWN IS LOCATED IN ATTIC AND SHOWN ON FLOOR PLANS FOR COORDINATION PURPOSES ONLY. SEE ATTIC AND SECTION VIEWS FOR ELEVATIONS AND LOCATIONS OF EQUIPMENT.

### # PLAN NOTES

1. PROVIDE 4-PIPE FAN COIL UNIT ABOVE SECOND GYPSUM BOARD CEILING. PROVIDE RETURN DUCT PLENUM TO MATCH DUCT INLET CONNECTION. PROVIDE TRANSITION OR TAP AS SHOWN AND ROUTE SUPPLY AND RETURN DUCTWORK DOWN THROUGH GYP CEILING TO LAY-IN CEILING GRILLES. PROVIDE DUCT MOUNTED BALANCING DAMPER BETWEEN GYP CEILING AND GRILLE.
2. PROVIDE VENTILATION DUCTWORK DOWN FROM DOAS MAIN DUCTWORK IN ATTIC, TO LAY-IN CEILING MOUNTED GRILLES AS SHOWN. SEE ATTIC PLANS FOR CONTINUATION. (TYPICAL)
3. PROVIDE EXHAUST DUCTWORK DOWN FROM DOAS MAIN DUCTWORK IN ATTIC. ROUTE TO VERTICAL EXHAUST DUCTWORK IN CHASE DOWN TO SIDEWALL MOUNTED GRILLE AS SHOWN. (TYPICAL)




B1 FIRST FLOOR PLAN - WEST WING - MECHANICAL HVAC
3/16" = 1'-0"



A5 KEY PLAN - WEST WING
NTS

GRAPHIC SCALE: 3/16"=1'-0"
6' 0 4' 10'

# MH104

03-09-26

NBFA NO: 2406

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DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND

MARINE CORPS BASE

CAMP LEJEUNE, NORTH CAROLINA

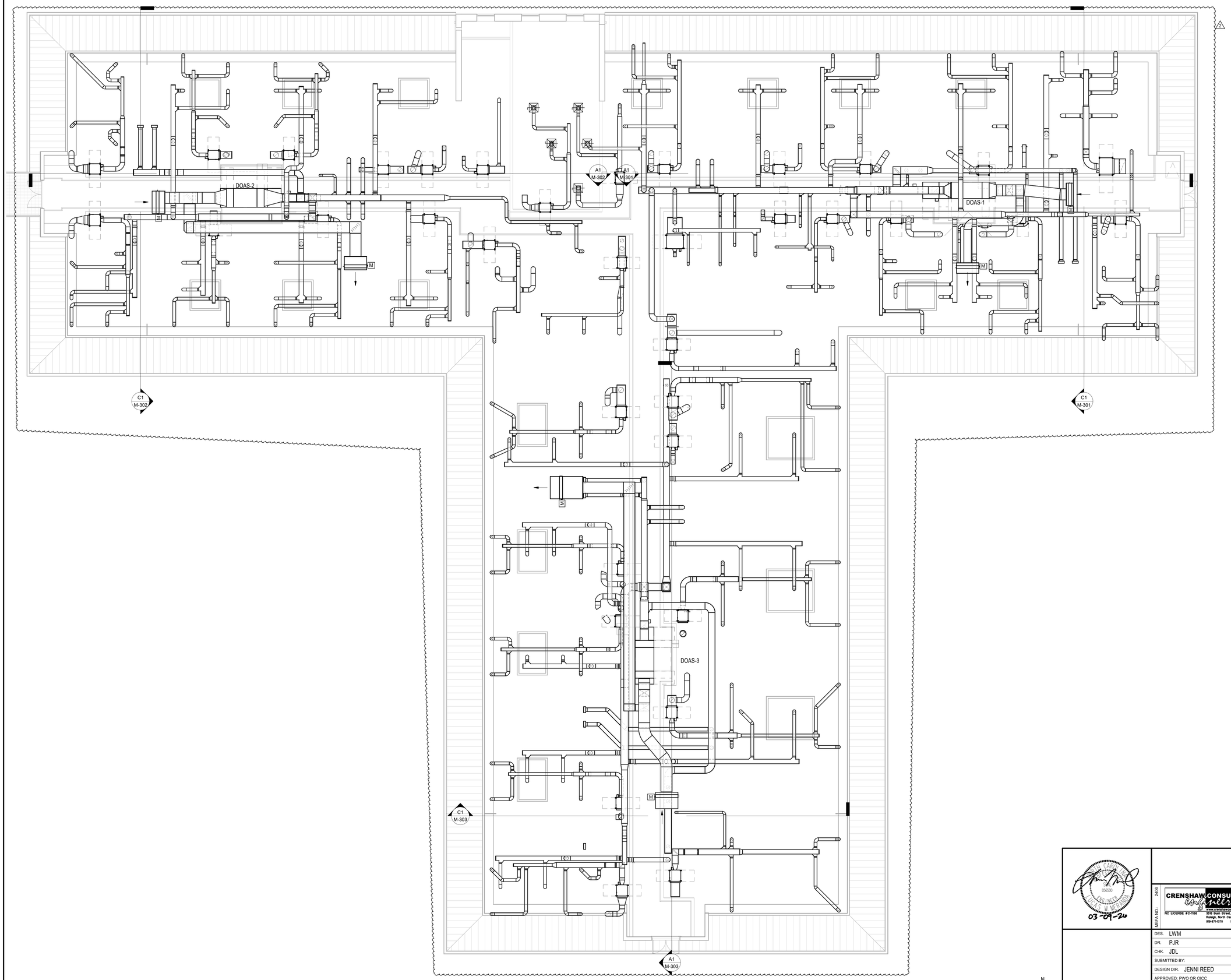
MECHANICAL REPAIRS
BUILDING TT113

FIRST FLOOR PLAN - WEST WING - MECHANICAL HVAC

DES. LWM
DR. PJR
CHK. JDL
SUBMITTED BY:
DESIGN DIR. JENNI REED
APPROVED: PWO OR OICC DATE
SATISFACTORY TO: DATE

| SIZE | CODE IDENT. NO. | NAVFAC DRAWING NO. |
|---|---|---|
| E1 | 80091 | 60042069 |
| | | CONSTR. CONTR. NO. N40085-24-F-5940 |
| SCALE AS NOTED | SPEC. | SHEET 62 OF 119 |

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| 2 | PPI REVISION 2 | 03.09.26 | |




GENERAL NOTES:

1. SEE SHEET M-001 FOR GENERAL NOTES, LEGEND & ABBREVIATIONS
2. PLAN DRAWINGS ARE DIAGRAMMATIC IN NATURE AND DO NO ILLUSTRATE SPECIFIC DUCT TAKE-OFF CONFIGURATIONS OR TAPS.

GRAPHIC SCALE: 1/8"=1'-0"

8' 0 4' 8' 16'

A1 ATTIC PLAN - OVERALL - MECHANICAL HVAC
1/8" = 1'-0"

N

MH105

03-09-26

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DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND

MARINE CORPS BASE

CAMP LEJEUNE, NORTH CAROLINA

DES. LWM
DR. PJR
CHK. JDL
SUBMITTED BY:
DESIGN DIR. JENNI REED
APPROVED: PWO OR OICC DATE
SATISFACTORY TO: DATE

MECHANICAL REPAIRS
BUILDING TT113

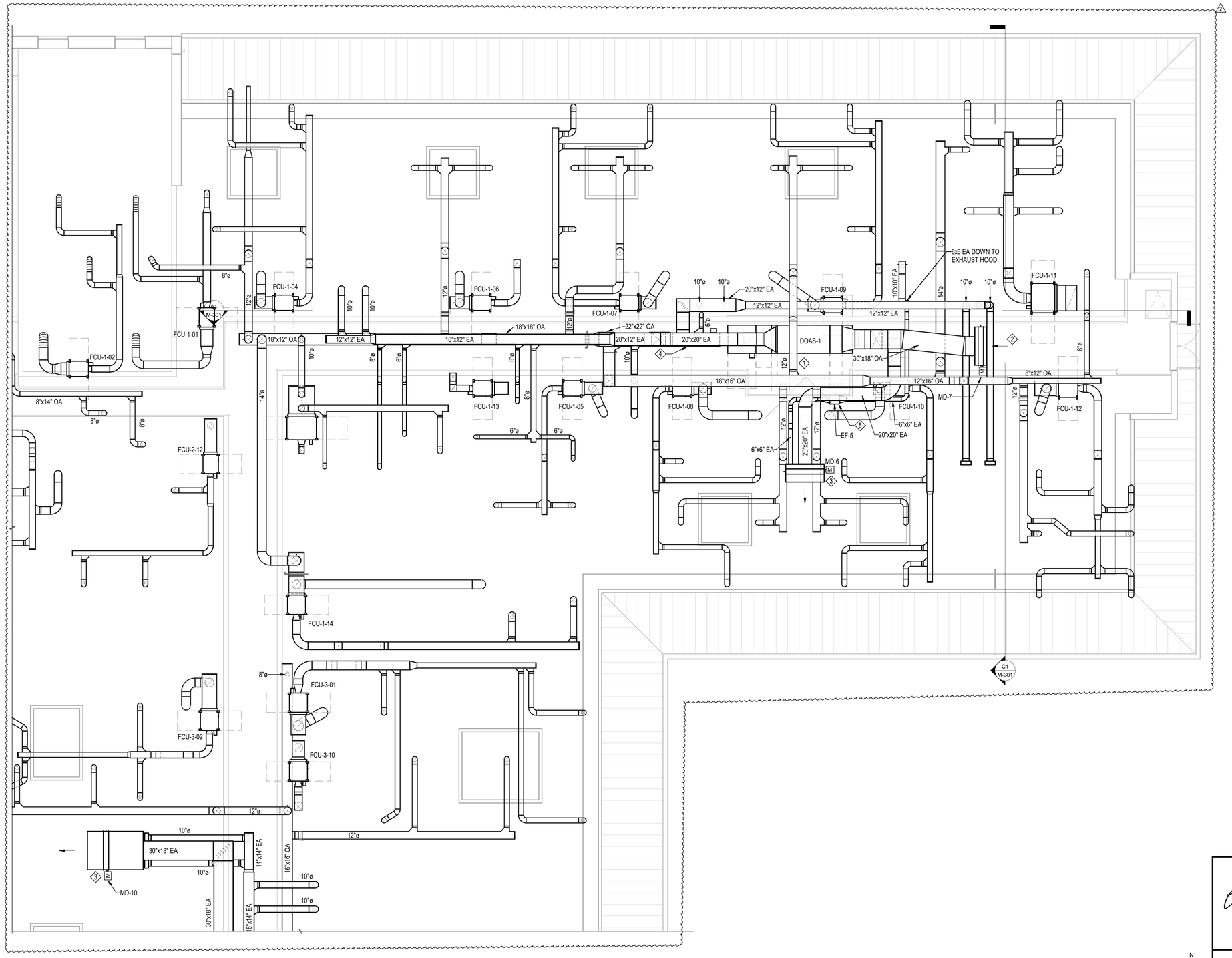
ATTIC PLAN - OVERALL - MECHANICAL HVAC

| SIZE | CODE IDENT. NO. | NAVFAC DRAWING NO. |
|---|---|---|
| E1 | 80091 | 60042070 |
| | | CONSTR. CONTR. NO. N40085-24-F-5940 |
| SCALE AS NOTED | SPEC. | SHEET 63 OF 119 |

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| 2 | PPI REVISION 2 | 03.09.26 | |
| | | | |
| | | | |

GENERAL NOTES:

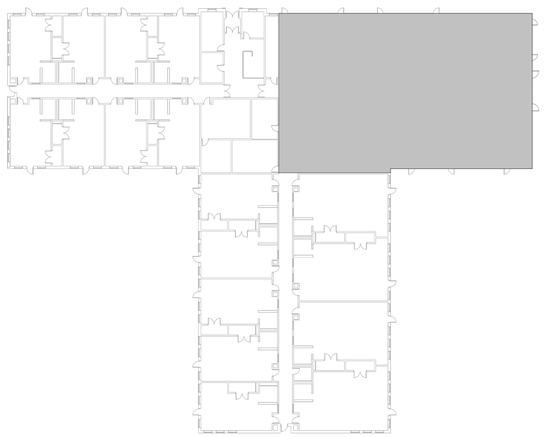
1. SEE SHEET M-001 FOR GENERAL NOTES, LEGEND & ABBREVIATIONS
2. PLAN DRAWINGS ARE DIAGRAMMATIC IN NATURE AND DO NO ILLUSTRATE SPECIFIC DUCT TAKE-OFF CONFIGURATIONS OR TAPS.




B1 ATTIC PLAN - EAST WING - MECHANICAL HVAC
3/16" = 1'-0"

PLAN NOTES

1. PROVIDE DOAS MOUNTED ON EQUIPMENT PLATFORM IN ATTIC. COORDINATE EXACT LOCATION ON PLATFORM WITH STRUCTURE AND PROVIDE MANUFACTURER RECOMMENDED CLEARANCES.
2. PROVIDE OUTSIDE AIR VENTILATION DUCTWORK FROM EXTERIOR WALL LOUVER WITH CLASS 1A MOTORIZED DAMPER TO DOAS OA INTAKE AS SHOWN. PROVIDE DUCT PLENUM BEHIND ARCHITECTURAL LOUVER, SIZE TO MATCH MOTORIZED DAMPER SIZE. SEE ARCHITECTURAL SHEETS FOR MORE INFORMATION ON LOUVER TYPE AND SIZE.
3. PROVIDE EXHAUST AIR DUCTWORK FROM EXTERIOR WALL LOUVER WITH CLASS 1A MOTORIZED DAMPER TO DOAS EXHAUST OUTLET AS SHOWN. PROVIDE DUCT PLENUM BEHIND ARCHITECTURAL LOUVER, SIZE TO MATCH MOTORIZED DAMPER SIZE. SEE ARCHITECTURAL SHEETS FOR MORE INFORMATION ON LOUVER TYPE AND SIZE.
4. PROVIDE OUTSIDE AIR VENTILATION DUCTWORK FROM DOAS OA OUTLET AND ROUTE AS SHOWN TO SPACES BELOW. SEE FLOOR PLANS FOR CONTINUATION
5. PROVIDE INLINE EXHAUST FAN AS SHOWN. INSTALL PER MANUFACTURER'S INSTALLATION AND CLEARANCE RECOMMENDATIONS. ROUTE DUCTWORK TO HOOD AND EXTERIOR LOUVER WITH CLASS 1A MOTORIZED DAMPER. SEE CONTROL DETAIL FOR SEQUENCE OF OPERATION.



A5 KEY PLAN - EAST WING
NTS

GRAPHIC SCALE: 3/16"=1'-0"
6' 0 4' 10'

MH106

03-09-26

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MBP/A NO: 24016

DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND

MARINE CORPS BASE

CAMP LEJEUNE, NORTH CAROLINA

DES. LWM
DR. PJR
CHK. JDL
SUBMITTED BY:
DESIGN DIR. JENNI REED
APPROVED: PWO OR OICC DATE
SATISFACTORY TO: DATE

MECHANICAL REPAIRS
BUILDING TT113

ATTIC PLAN - EAST WING - MECHANICAL HVAC

SIZE E1 | CODE IDENT. NO. 80091 | NAVFAC DRAWING NO. 60042071
CONSTR. CONTR. NO. N40085-24-F-5940
SCALE AS NOTED | SPEC. | SHEET 64 OF 119

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| 2 | PPI REVISION 2 | 03.09.26 | |
| | | | |
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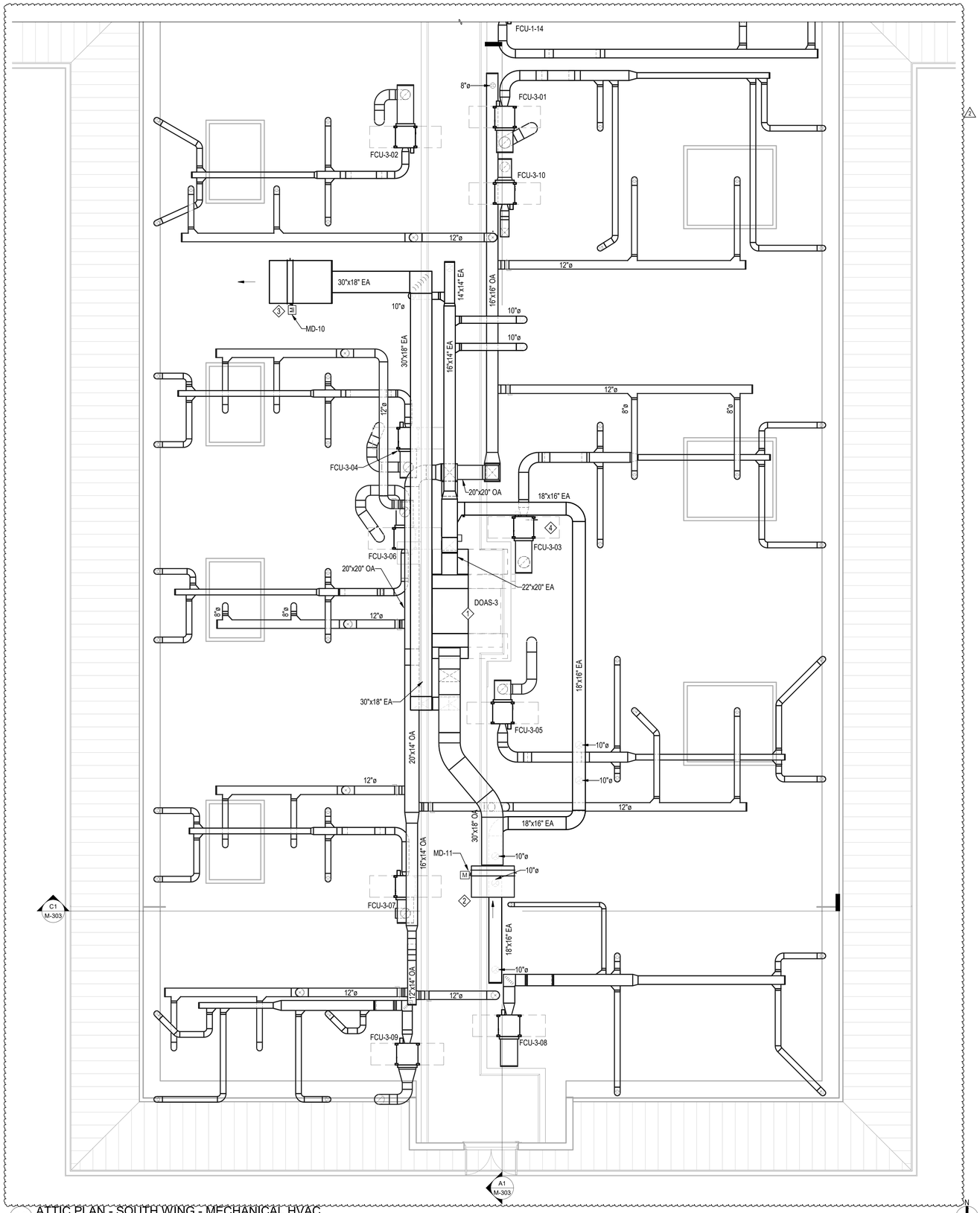



A2 ATTIC PLAN - SOUTH WING - MECHANICAL HVAC
3/16" = 1'-0"

GENERAL NOTES:

1. SEE SHEET M-001 FOR GENERAL NOTES, LEGEND & ABBREVIATIONS
2. PLAN DRAWINGS ARE DIAGRAMMATIC IN NATURE AND DO NO ILLUSTRATE SPECIFIC DUCT TAKE-OFF CONFIGURATIONS OR TAPS.

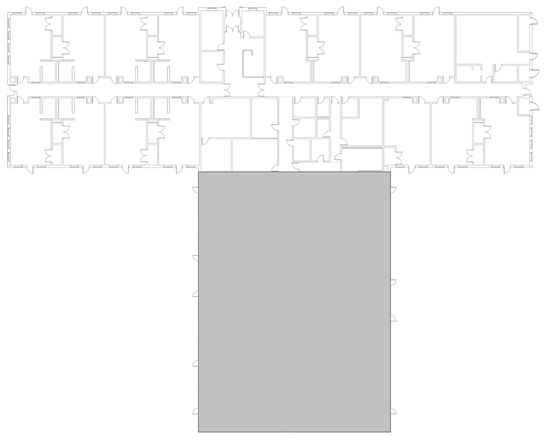
PLAN NOTES

1 PROVIDE DOAS MOUNTED ON EQUIPMENT PLATFORM IN ATTIC. COORDINATE EXACT LOCATION ON PLATFORM WITH STRUCTURE AND PROVIDE MANUFACTURER RECOMMENDED CLEARANCES.

2 PROVIDE OUTSIDE AIR VENTILATION DUCTWORK FROM EXTERIOR WALL LOUVER WITH CLASS 1A MOTORIZED DAMPER TO DOAS OA INTAKE AS SHOWN. PROVIDE DUCT PLENUM BEHIND ARCHITECTURAL LOUVER, SIZE TO MATCH MOTORIZED DAMPER SIZE. SEE ARCHITECTURAL SHEETS FOR MORE INFORMATION ON LOUVER TYPE AND SIZE.

3 PROVIDE EXHAUST AIR DUCTWORK FROM EXTERIOR WALL LOUVER WITH CLASS 1A MOTORIZED DAMPER TO DOAS EXHAUST OUTLET AS SHOWN. PROVIDE DUCT PLENUM BEHIND ARCHITECTURAL LOUVER, SIZE TO MATCH MOTORIZED DAMPER SIZE. SEE ARCHITECTURAL SHEETS FOR MORE INFORMATION ON LOUVER TYPE AND SIZE.

4 PROVIDE EXHAUST AIR DUCTWORK FROM DOAS EA INLET AND ROUTE AS SHOWN TO SPACES BELOW. SEE FLOOR PLANS FOR CONTINUATION.



A5 KEY PLAN - SOUTH WING
NTS

GRAPHIC SCALE: 3/16"=1'-0"
6' 0 4' 10'

MH107

03-09-26

MBFA NO: 2406

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Raleigh, North Carolina 27609
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DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND

MARINE CORPS BASE

CAMP LEJEUNE, NORTH CAROLINA

DES. LWM
DR. PJR
CHK. JDL
SUBMITTED BY:
DESIGN DIR. JENNI REED
APPROVED: PWO OR OICC DATE
SATISFACTORY TO: DATE

MECHANICAL REPAIRS
BUILDING TT113

ATTIC PLAN - SOUTH WING - MECHANICAL HVAC

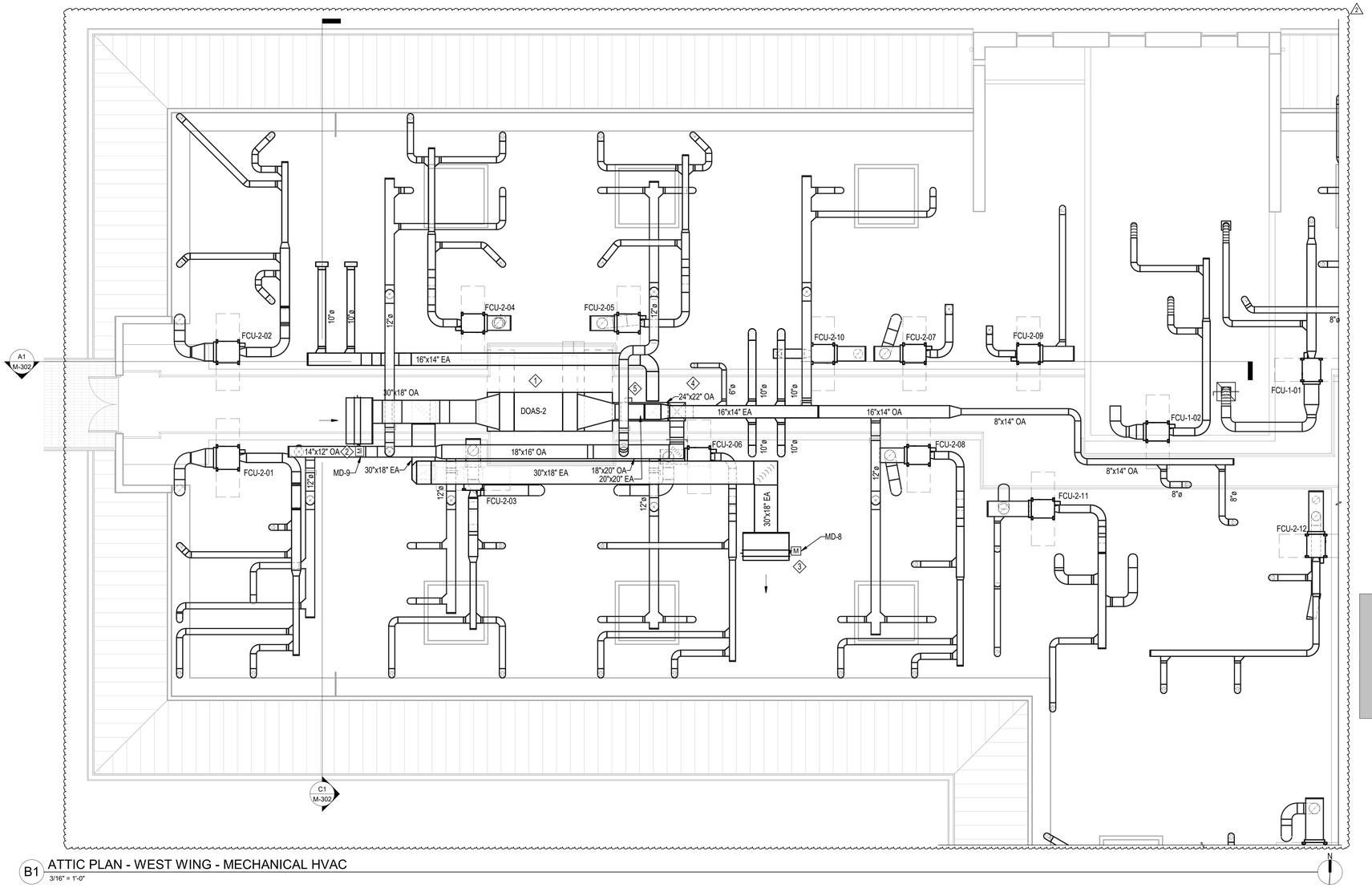
| SIZE | CODE IDENT. NO. | NAVFAC DRAWING NO. |
|---|---|---|
| E1 | 80091 | 60042072 |
| | | CONSTR. CONTR. NO. N40085-24-F-5940 |

SCALE AS NOTED SPEC. SHEET 65 OF 119

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| 2 | PPI REVISION 2 | 03.09.26 | |
| | | | |
| | | | |

GENERAL NOTES:

1. SEE SHEET M-001 FOR GENERAL NOTES, LEGEND & ABBREVIATIONS
2. PLAN DRAWINGS ARE DIAGRAMMATIC IN NATURE AND DO NO ILLUSTRATE SPECIFIC DUCT TAKE-OFF CONFIGURATIONS OR TAPS.




B1 ATTIC PLAN - WEST WING - MECHANICAL HVAC
3/16" = 1'-0"

PLAN NOTES

1 PROVIDE DOAS MOUNTED ON EQUIPMENT PLATFORM IN ATTIC. COORDINATE EXACT LOCATION ON PLATFORM WITH STRUCTURE AND PROVIDE MANUFACTURER RECOMMENDED CLEARANCES.

2 PROVIDE OUTSIDE AIR VENTILATION DUCTWORK FROM EXTERIOR WALL LOUVER WITH CLASS 1A MOTORIZED DAMPER TO DOAS OA INTAKE AS SHOWN. PROVIDE DUCT PLENUM BEHIND ARCHITECTURAL LOUVER, SIZE TO MATCH MOTORIZED DAMPER SIZE. SEE ARCHITECTURAL SHEETS FOR MORE INFORMATION ON LOUVER TYPE AND SIZE.

3 PROVIDE EXHAUST AIR DUCTWORK FROM EXTERIOR WALL LOUVER WITH CLASS 1A MOTORIZED DAMPER TO DOAS EXHAUST OUTLET AS SHOWN. PROVIDE DUCT PLENUM BEHIND ARCHITECTURAL LOUVER, SIZE TO MATCH MOTORIZED DAMPER SIZE. SEE ARCHITECTURAL SHEETS FOR MORE INFORMATION ON LOUVER TYPE AND SIZE.

4 PROVIDE OUTSIDE AIR VENTILATION DUCTWORK FROM DOAS OA OUTLET AND ROUTE AS SHOWN TO SPACES BELOW. SEE FLOOR PLANS FOR CONTINUATION

5 PROVIDE EXHAUST AIR DUCTWORK FROM DOAS EA INLET AND ROUTE AS SHOWN TO SPACES BELOW. SEE FLOOR PLANS FOR CONTINUATION.

A5 KEY PLAN - WEST WING
NTS

GRAPHIC SCALE: 3/16"=1'-0"
6' 0 4' 10'

MH108

03-09-26

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DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND

MARINE CORPS BASE

CAMP LEJEUNE, NORTH CAROLINA

DES. LWM
DR. PJR
CHK. JDL
SUBMITTED BY:
DESIGN DIR. JENNI REED
APPROVED: PWO OR OICC DATE
SATISFACTORY TO: DATE

MECHANICAL REPAIRS
BUILDING TT113

ATTIC PLAN - WEST WING - MECHANICAL HVAC

| SIZE | CODE IDENT. NO. | NAVFAC DRAWING NO. |
|---|---|---|
| E1 | 80091 | 60042073 |
| | | CONSTR. CONTR. NO. N40085-24-F-5940 |

SCALE AS NOTED SPEC. SHEET 66 OF 119

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| 2 | PPI REVISION 2 | 03.09.26 | |
| | | | |
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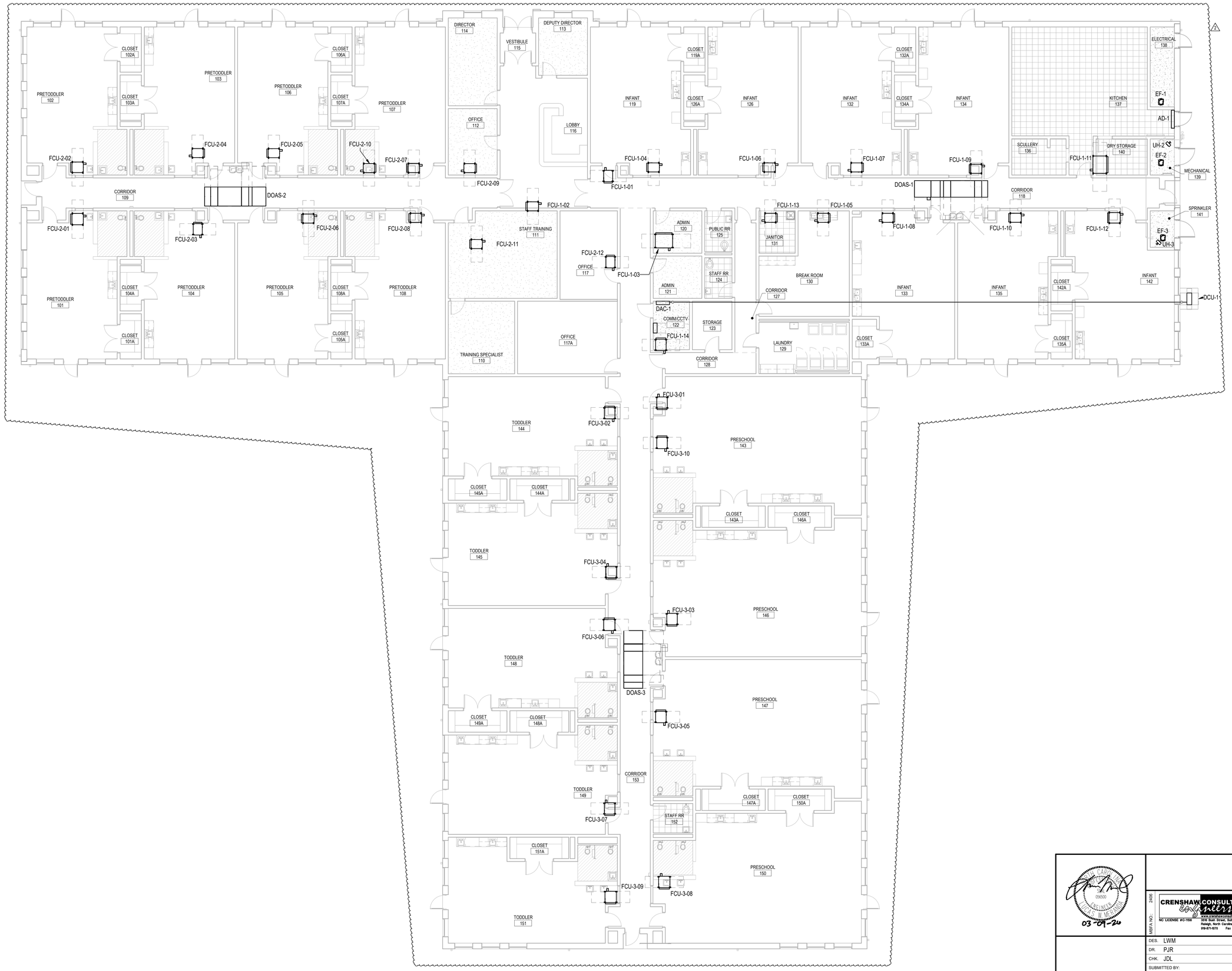


## GENERAL NOTES:

1. SEE SHEET M-001 FOR GENERAL NOTES, LEGEND & ABBREVIATIONS
2. SEE CHILLED AND HOT WATER SYSTEM DIAGRAMS FOR PIPE SIZING AND MORE INFORMATION.
3. FAN COIL UNITS AND ASSOCIATED PIPING SHOWN IS LOCATED IN ATTIC AND SHOWN ON FLOOR PLANS FOR COORDINATION PURPOSES ONLY. SEE SECTION VIEWS FOR ELEVATIONS AND LOCATIONS OF EQUIPMENT.

GRAPHIC SCALE: 1/8"=1'-0"
8' 0 4' 8' 16'

A1 FIRST FLOOR PLAN - OVERALL - MECHANICAL PIPING
1/8" = 1'-0"

MP101

03-09-26

NBFA NO: 2406

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DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND

MARINE CORPS BASE

CAMP LEJEUNE, NORTH CAROLINA

DES. LWM
DR. PJR
CHK. JDL
SUBMITTED BY:
DESIGN DIR. JENNI REED
APPROVED: PWO OR OICC DATE
SATISFACTORY TO: DATE

MECHANICAL REPAIRS
BUILDING TT113

FIRST FLOOR PLAN - OVERALL - MECHANICAL PIPING

| SIZE | CODE IDENT. NO. | NAVFAC DRAWING NO. |
|---|---|---|
| E1 | 80091 | 60042074 |
| | | CONSTR. CONTR. NO. N40085-24-F-5940 |
| SCALE AS NOTED | SPEC. | SHEET 67 OF 119 |

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| 2 | PPI REVISION 2 | 03.09.26 | |
| | | | |
| | | | |

GENERAL NOTES:

1. SEE SHEET M-001 FOR GENERAL NOTES, LEGEND & ABBREVIATIONS
2. SEE CHILLED AND HOT WATER SYSTEM DIAGRAMS FOR PIPE SIZING AND MORE INFORMATION.
3. FAN COIL UNITS AND ASSOCIATED PIPING SHOWN IS LOCATED IN ATTIC AND SHOWN ON FLOOR PLANS FOR COORDINATION PURPOSES ONLY. SEE SECTION VIEWS FOR ELEVATIONS AND LOCATIONS OF EQUIPMENT.

PLAN NOTES

1 PROVIDE 4-PIPE FAN COIL UNIT (FCU) ABOVE SECOND GYPSUM BOARD CEILING. ROUTE CHILLED AND HOT WATER PIPING TO UNIT AND PROVIDE MANUFACTURER RECOMMENDED CLEARANCES FOR SERVICE OF COILS AND VALVING. PROVIDE FCU WITH SECONDARY CONDENSATE DRAIN PAN. SEE FAN COIL DETAIL FOR MORE INFORMATION.

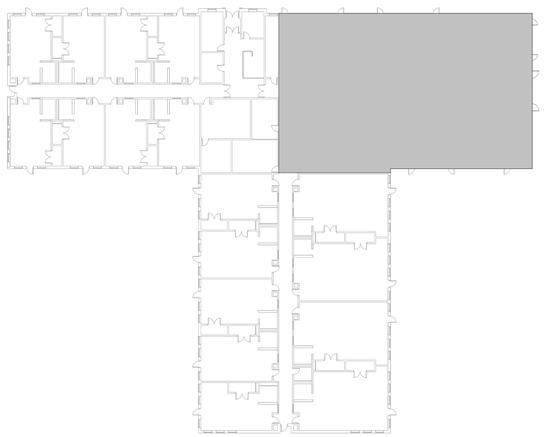
2 PROVIDE WALL MOUNTED THERMOSTAT MOUNTED AT 54" AFF.

3 PROVIDE CHILLED AND HOT WATER PIPING UP FROM UNDERGROUND AND ROUTE TO EQUIPMENT IN ATTIC. SEE ATTIC PLANS FOR CONTINUATION.

4 PROVIDE HOT WATER UNIT HEATERS SUSPENDED FROM STRUCTURE. COORDINATE EXACT LOCATION WITH EQUIPMENT IN SPACE.

5 PROVIDE WALL MOUNTED DUCTLESS SPLIT AIR HANDLER ABOVE DOOR. PROVIDE AIR HANDLER WITH CONDENSATE PUMP AND ROUTE CONDENSATE TO NEAREST DRAIN. ROUTE REFRIGERANT PIPING IN MOST DIRECT ROUTE POSSIBLE TO EXTERIOR CONDENSING UNIT MOUNTED ON EQUIPMENT PAD ON GRADE..

B1 FIRST FLOOR PLAN - EAST WING - MECHANICAL PIPING
3/16" = 1'-0"



A5 KEY PLAN - EAST WING
NTS

GRAPHIC SCALE: 3/16"=1'-0"
6' 0 4' 10'

03-09-26

MP102

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MBFA NO.: 24/06

DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND

MARINE CORPS BASE

CAMP LEJEUNE, NORTH CAROLINA

DES. LWM
DR. PJR
CHK. JDL
SUBMITTED BY:
DESIGN DIR. JENNI REED
APPROVED: PWO OR OICC DATE
SATISFACTORY TO: DATE

MECHANICAL REPAIRS
BUILDING TT113

FIRST FLOOR PLAN - EAST WING - MECHANICAL PIPING

| SIZE | CODE IDENT. NO. | NAVFAC DRAWING NO. |
|---|---|---|
| E1 | 80091 | 60042075 |
| | | CONSTR. CONTR. NO. N40085-24-F-5940 |

SCALE AS NOTED SPEC. SHEET 68 OF 119

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| 2 | PPI REVISION 2 | 03.09.26 | |
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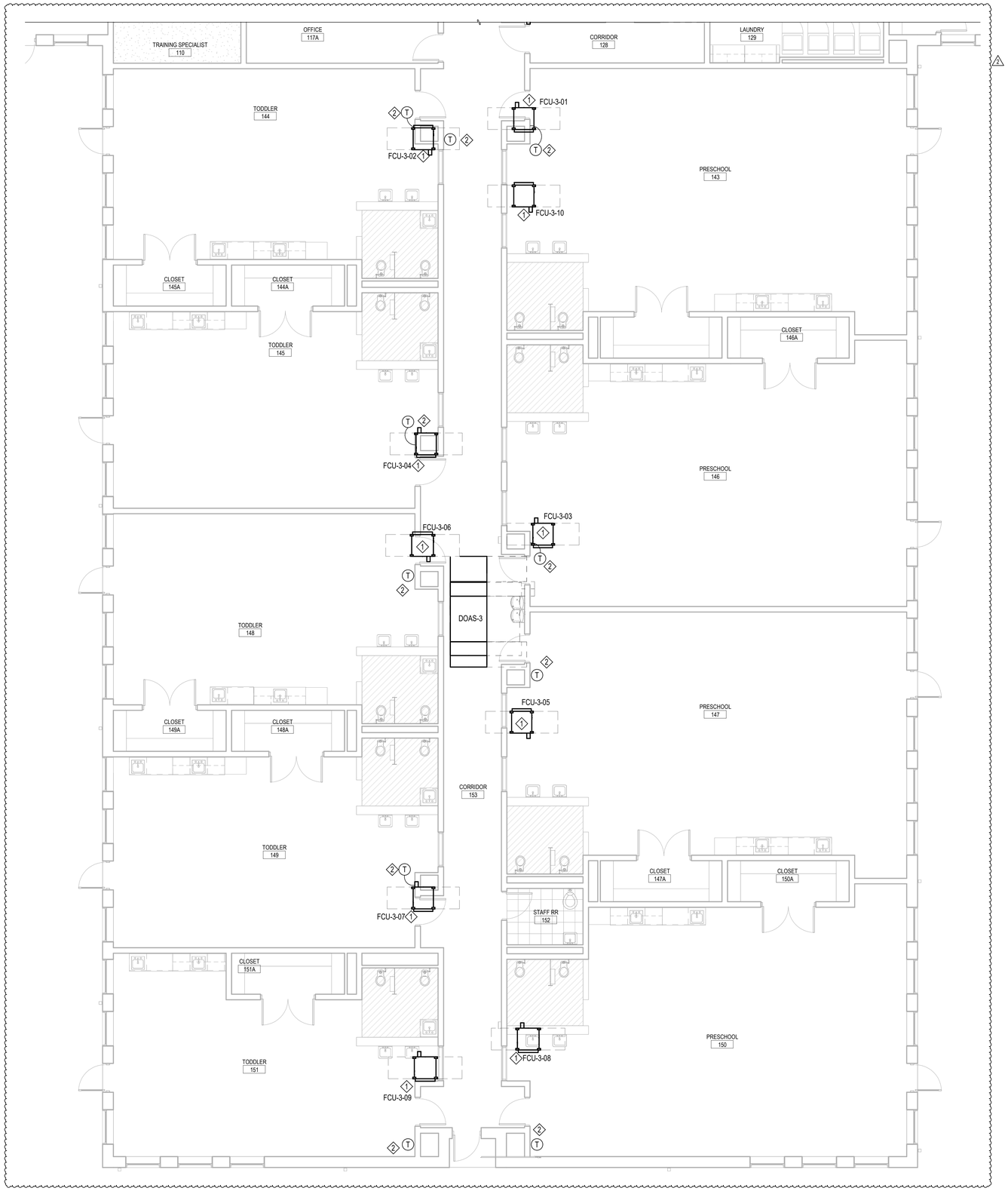



A2 FIRST FLOOR PLAN - SOUTH WING - MECHANICAL PIPING
3/16" = 1'-0"

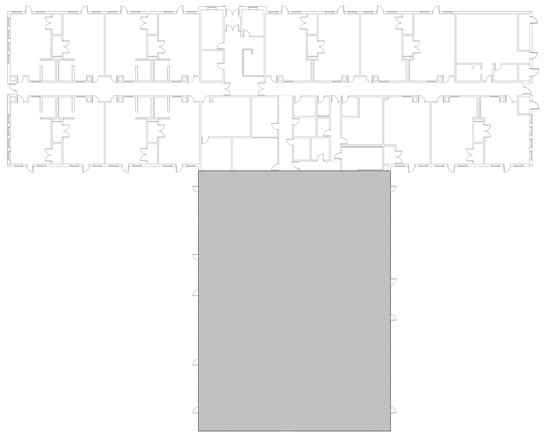
GENERAL NOTES:

1. SEE SHEET M-001 FOR GENERAL NOTES, LEGEND & ABBREVIATIONS
2. SEE CHILLED AND HOT WATER SYSTEM DIAGRAMS FOR PIPE SIZING AND MORE INFORMATION.
3. FAN COIL UNITS AND ASSOCIATED PIPING SHOWN IS LOCATED IN ATTIC AND SHOWN ON FLOOR PLANS FOR COORDINATION PURPOSES ONLY. SEE SECTION VIEWS FOR ELEVATIONS AND LOCATIONS OF EQUIPMENT.

PLAN NOTES

1 PROVIDE 4-PIPE FAN COIL UNIT (FCU) ABOVE SECOND GYPSUM BOARD CEILING. ROUTE CHILLED AND HOT WATER PIPING TO UNIT AND PROVIDE MANUFACTURER RECOMMENDED CLEARANCES FOR SERVICE OF COILS AND VALVING. PROVIDE FCU WITH SECONDARY CONDENSATE DRAIN PAN. SEE FAN COIL DETAIL FOR MORE INFORMATION.

2 PROVIDE WALL MOUNTED THERMOSTAT MOUNTED AT 54" AFF.



A5 KEY PLAN - SOUTH WING
NTS

GRAPHIC SCALE: 3/16"=1'-0"
6' 0 4' 10'

MP103

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CAMP LEJEUNE, NORTH CAROLINA

DES. LWM
DR. PJR
CHK. JDL
SUBMITTED BY:
DESIGN DIR. JENNI REED
APPROVED: PWO OR OICC DATE
SATISFACTORY TO: DATE

MECHANICAL REPAIRS
BUILDING TT113

FIRST FLOOR PLAN - SOUTH WING - MECHANICAL PIPING

SIZE E1 | CODE IDENT. NO. 80091 | NAVFAC DRAWING NO. 60042076
CONSTR. CONTR. NO. N40085-24-F-5940
SCALE AS NOTED | SPEC. | SHEET 69 OF 119

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| 2 | PPI REVISION 2 | 03.09.26 | |
| | | | |
| | | | |

GENERAL NOTES:

1. SEE SHEET M-001 FOR GENERAL NOTES, LEGEND & ABBREVIATIONS
2. SEE CHILLED AND HOT WATER SYSTEM DIAGRAMS FOR PIPE SIZING AND MORE INFORMATION.
3. FAN COIL UNITS AND ASSOCIATED PIPING SHOWN IS LOCATED IN ATTIC AND SHOWN ON FLOOR PLANS FOR COORDINATION PURPOSES ONLY. SEE SECTION VIEWS FOR ELEVATIONS AND LOCATIONS OF EQUIPMENT.

PLAN NOTES

1 PROVIDE 4-PIPE FAN COIL UNIT (FCU) ABOVE SECOND GYPSUM BOARD CEILING. ROUTE CHILLED AND HOT WATER PIPING TO UNIT AND PROVIDE MANUFACTURER RECOMMENDED CLEARANCES FOR SERVICE OF COILS AND VALVING. PROVIDE FCU WITH SECONDARY CONDENSATE DRAIN PAN. SEE FAN COIL DETAIL FOR MORE INFORMATION.

2 PROVIDE WALL MOUNTED THERMOSTAT MOUNTED AT 54" AFF.

B1 FIRST FLOOR PLAN - WEST WING - MECHANICAL PIPING
3/16" = 1'-0"



A5 KEY PLAN - WEST WING
NTS

GRAPHIC SCALE: 3/16"=1'-0"
6' 0 4' 10'

MP104

03-09-26

MBFA NO: 2406

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MECHANICAL REPAIRS
BUILDING TT113

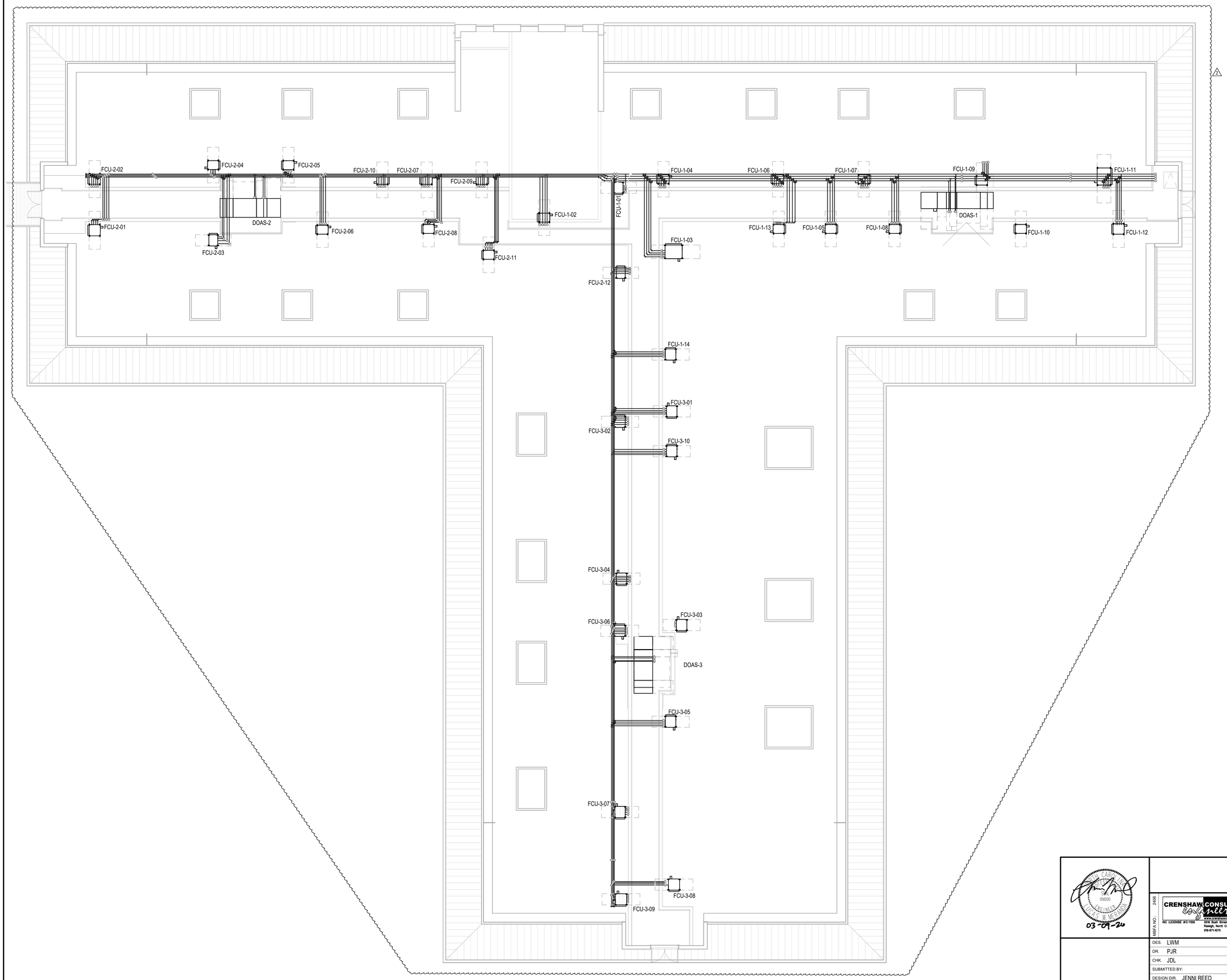
FIRST FLOOR PLAN - WEST WING - MECHANICAL PIPING

SIZE E1 | CODE IDENT. NO. 80091 | NAVFAC DRAWING NO. 60042077
CONSTR. CONTR. NO. N40085-24-F-5940
SCALE AS NOTED | SPEC. | SHEET 70 OF 119

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| 2 | PPI REVISION 2 | 03.09.26 | |

GENERAL NOTES:

1. SEE SHEET M-001 FOR GENERAL NOTES, LEGEND & ABBREVIATIONS
2. SEE CHILLED AND HOT WATER SYSTEM DIAGRAMS FOR PIPE SIZING AND MORE INFORMATION.
3. FAN COIL UNITS AND ASSOCIATED PIPING SHOWN IS LOCATED IN ATTIC AND SHOWN ON FLOOR PLANS FOR COORDINATION PURPOSES ONLY. SEE SECTION VIEWS FOR ELEVATIONS AND LOCATIONS OF EQUIPMENT.




A1 ATTIC PLAN - OVERALL - MECHANICAL PIPING
1/8" = 1'-0"

GRAPHIC SCALE: 1/8"=1'-0"
8' 0 4' 8' 16'

MP105

03-09-26

MBFA NO: 2406

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MARINE CORPS BASE

CAMP LEJEUNE, NORTH CAROLINA

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CHK. JDL
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SATISFACTORY TO: DATE

MECHANICAL REPAIRS
BUILDING TT113

ATTIC PLAN - OVERALL - MECHANICAL PIPING

| SIZE | CODE IDENT. NO. | NAVFAC DRAWING NO. |
|---|---|---|
| E1 | 80091 | 60042078 |
| | | CONSTR. CONTR. NO. N40085-24-F-5940 |

SCALE AS NOTED SPEC. SHEET 71 OF 119

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| 2 | PPI REVISION 2 | 03.09.26 | |
| | | | |
| | | | |

GENERAL NOTES:

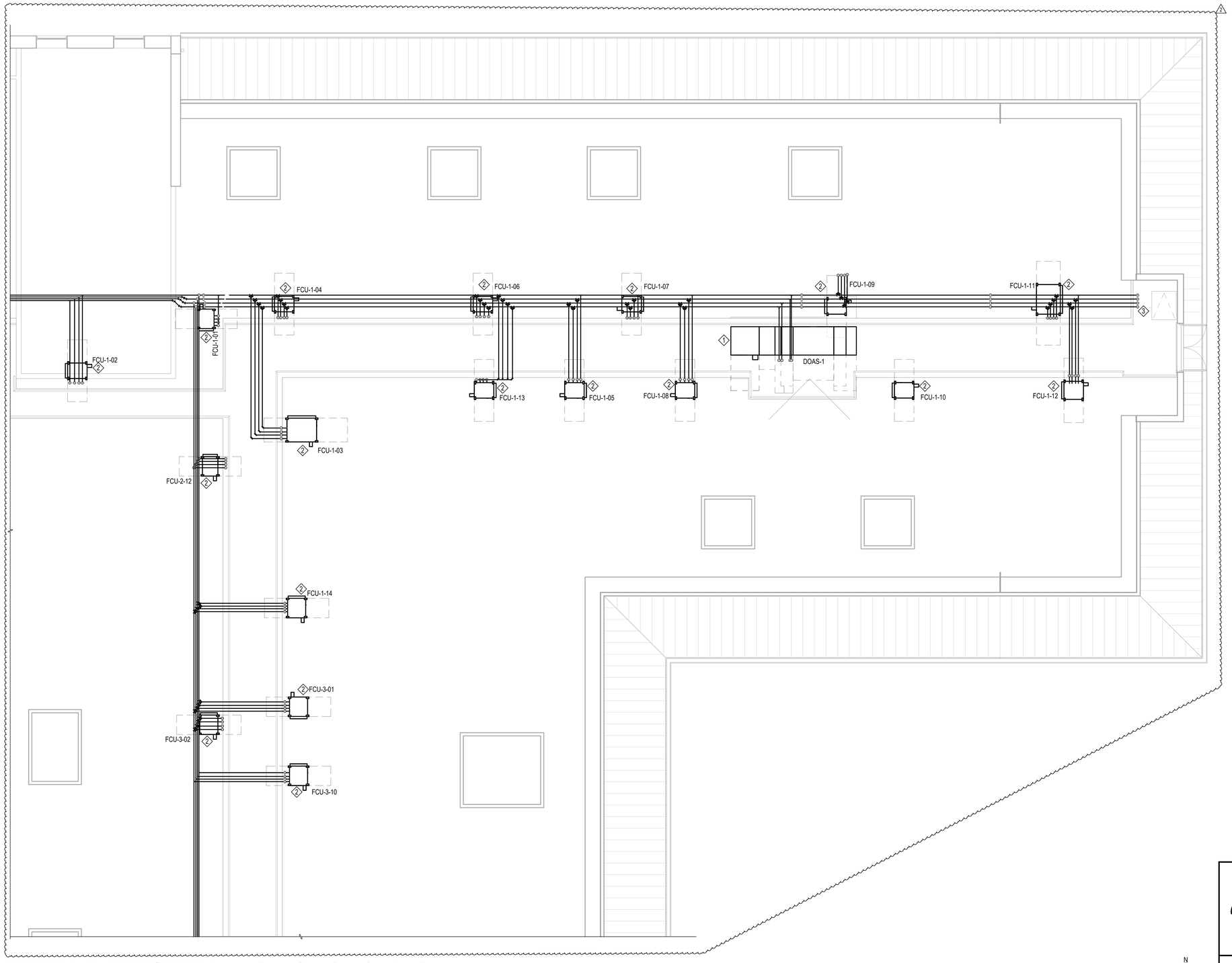
1. SEE SHEET M-001 FOR GENERAL NOTES, LEGEND & ABBREVIATIONS
2. SEE CHILLED AND HOT WATER SYSTEM DIAGRAMS FOR PIPE SIZING AND MORE INFORMATION.
3. FAN COIL UNITS AND ASSOCIATED PIPING SHOWN IS LOCATED IN ATTIC AND SHOWN ON FLOOR PLANS FOR COORDINATION PURPOSES ONLY. SEE SECTION VIEWS FOR ELEVATIONS AND LOCATIONS OF EQUIPMENT.

PLAN NOTES

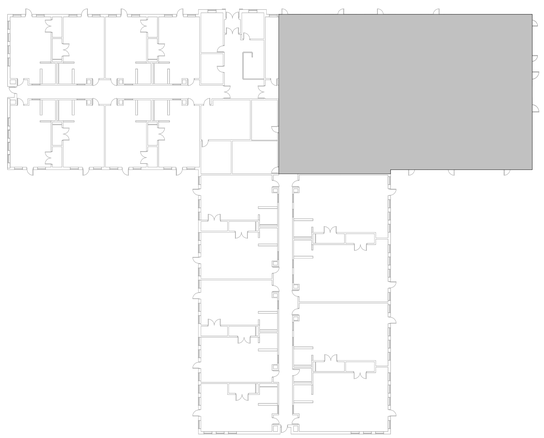
1 PROVIDE DOAS MOUNTED ON EQUIPMENT PLATFORM IN ATTIC. ROUTE CHILLED AND HOT WATER PIPING TO UNIT AND PROVIDE MANUFACTURER RECOMMENDED CLEARANCES FOR SERVICE OF COILS AND VALVING.

2 PROVIDE 4-PIPE FAN COIL UNIT (FCU) ABOVE SECOND GYPSUM BOARD CEILING. UNIT TO BE INSTALLED AT ELEVATION AND ORIENTATION TO ALLOW FOR MAINTENANCE ACCESS FROM CATWALK. PROVIDE CHILLED AND HOT WATER PIPING TO 4-PIPE FAN COIL UNIT (FCU). INSTALL PER MANUFACTURER INSTALLATION INSTRUCTIONS AND PROVIDE RECOMMENDED CLEARANCES FOR SERVICE OF COILS AND VALVING. FCU TO BE PROVIDED WITH SECONDARY CONDENSATE DRAIN PAN. SEE FLOOR PLANS FOR CONTINUATION.

3 PROVIDE CHILLED AND HOT WATER PIPING UP FROM LEVEL BELOW AND ROUTE TO EQUIPMENT IN ATTIC AS SHOWN. COORDINATE EXACT ROUTING WITH OTHER EQUIPMENT, DUCTWORK AND STRUCTURE IN ATTIC.




B1 ATTIC PLAN - EAST WING - MECHANICAL PIPING
3/16" = 1'-0"



A5 KEY PLAN - EAST WING
NTS

GRAPHIC SCALE: 3/16"=1'-0"
6' 0 4' 10'

MP106

03-09-26

MBFA NO.: 2406
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DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND

MARINE CORPS BASE

CAMP LEJEUNE, NORTH CAROLINA

DES. LWM
DR. PJR
CHK. JDL
SUBMITTED BY:
DESIGN DIR. JENNI REED
APPROVED: PWO OR OICC DATE
SATISFACTORY TO: DATE

MECHANICAL REPAIRS
BUILDING TT113

ATTIC PLAN - EAST WING - MECHANICAL PIPING

| SIZE | CODE IDENT. NO. | NAVFAC DRAWING NO. |
|---|---|---|
| E1 | 80091 | 60042079 |
| | | CONSTR. CONTR. NO. N40085-24-F-5940 |

SCALE AS NOTED SPEC. SHEET 72 OF 119

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| 2 | PPI REVISION 2 | 03.09.26 | |
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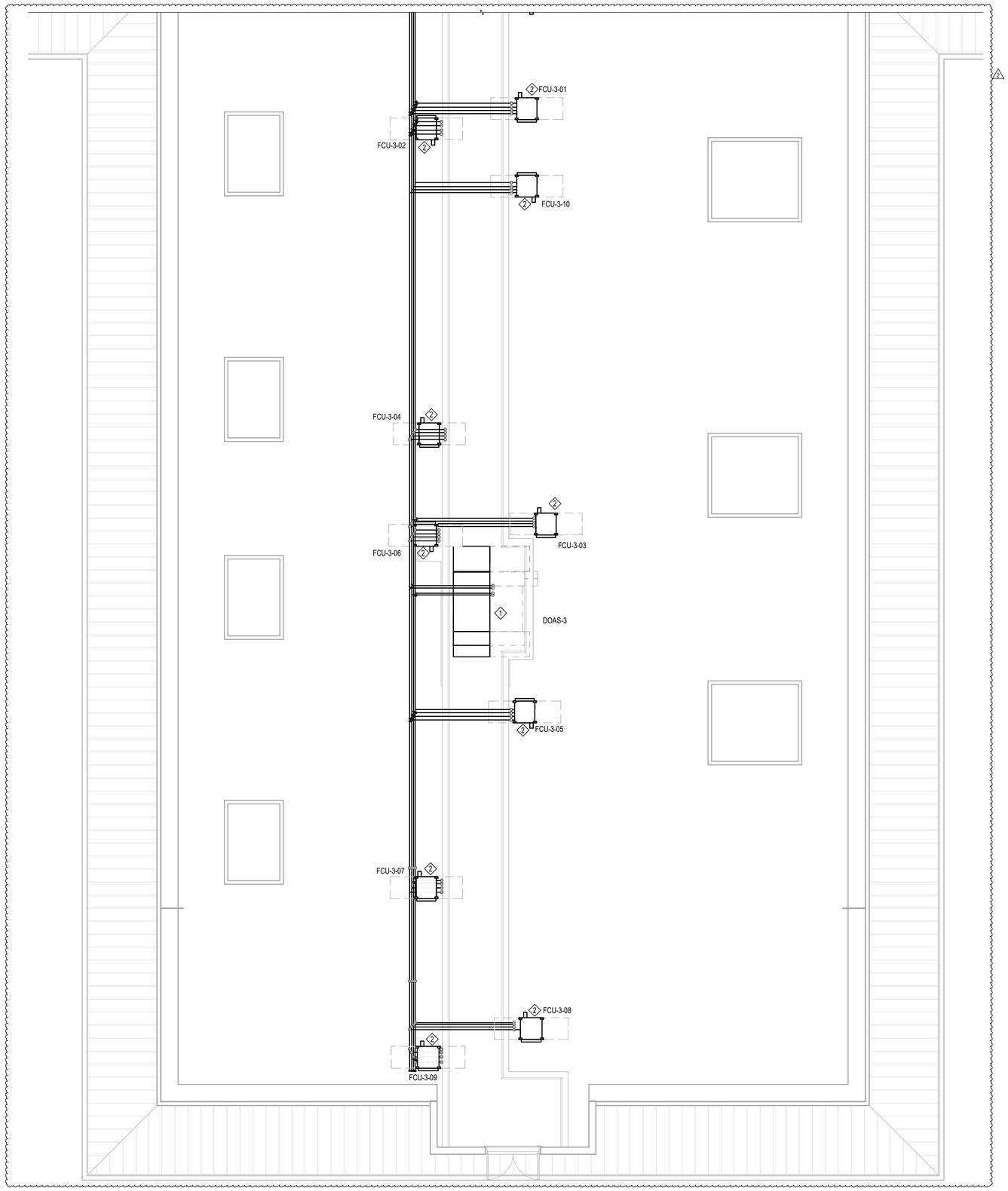



A2 ATTIC PLAN - SOUTH WING - MECHANICAL PIPING
3/16" = 1'-0"

N

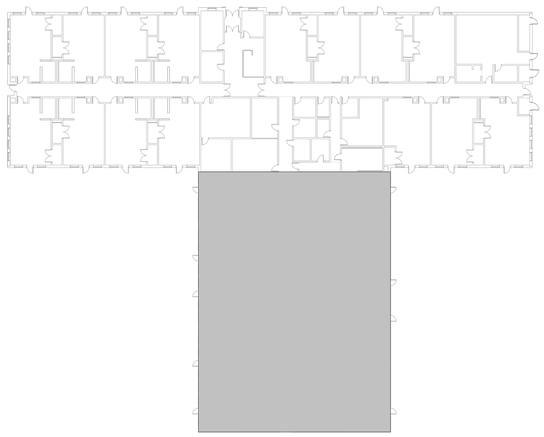
GENERAL NOTES:

1. SEE SHEET M-001 FOR GENERAL NOTES, LEGEND & ABBREVIATIONS
2. SEE CHILLED AND HOT WATER SYSTEM DIAGRAMS FOR PIPE SIZING AND MORE INFORMATION.
3. FAN COIL UNITS AND ASSOCIATED PIPING SHOWN IS LOCATED IN ATTIC AND SHOWN ON FLOOR PLANS FOR COORDINATION PURPOSES ONLY. SEE SECTION VIEWS FOR ELEVATIONS AND LOCATIONS OF EQUIPMENT.

# PLAN NOTES

1 PROVIDE DOAS MOUNTED ON EQUIPMENT PLATFORM IN ATTIC. ROUTE CHILLED AND HOT WATER PIPING TO UNIT AND PROVIDE MANUFACTURER RECOMMENDED CLEARANCES FOR SERVICE OF COILS AND VALVING.

2 PROVIDE 4-PIPE FAN COIL UNIT (FCU) ABOVE SECOND GYPSUM BOARD CEILING. UNIT TO BE INSTALLED AT ELEVATION AND ORIENTATION TO ALLOW FOR MAINTENANCE ACCESS FROM CATWALK. PROVIDE CHILLED AND HOT WATER PIPING TO 4-PIPE FAN COIL UNIT (FCU). INSTALL PER MANUFACTURER INSTALLATION INSTRUCTIONS AND PROVIDE RECOMMENDED CLEARANCES FOR SERVICE OF COILS AND VALVING. FCU TO BE PROVIDED WITH SECONDARY CONDENSATE DRAIN PAN. SEE FLOOR PLANS FOR CONTINUATION.



A5 KEY PLAN - SOUTH WING
NTS

GRAPHIC SCALE: 3/16"=1'-0"
6' 0 4' 10'

MP107

03-09-26

MBFA NO.: 2406

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DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND

MARINE CORPS BASE

CAMP LEJEUNE, NORTH CAROLINA

DES. LWM
DR. PJR
CHK. JDL
SUBMITTED BY:
DESIGN DIR. JENNI REED
APPROVED: PWO OR OICC DATE
SATISFACTORY TO: DATE

MECHANICAL REPAIRS
BUILDING TT113

ATTIC PLAN - SOUTH WING - MECHANICAL PIPING

| SIZE | CODE IDENT. NO. | NAVFAC DRAWING NO. |
|---|---|---|
| E1 | 80091 | 60042080 |
| | | CONSTR. CONTR. NO. N40085-24-F-5940 |

SCALE AS NOTED SPEC. SHEET 73 OF 119

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| 2 | PPI REVISION 2 | 03.09.26 | |
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| | | | |

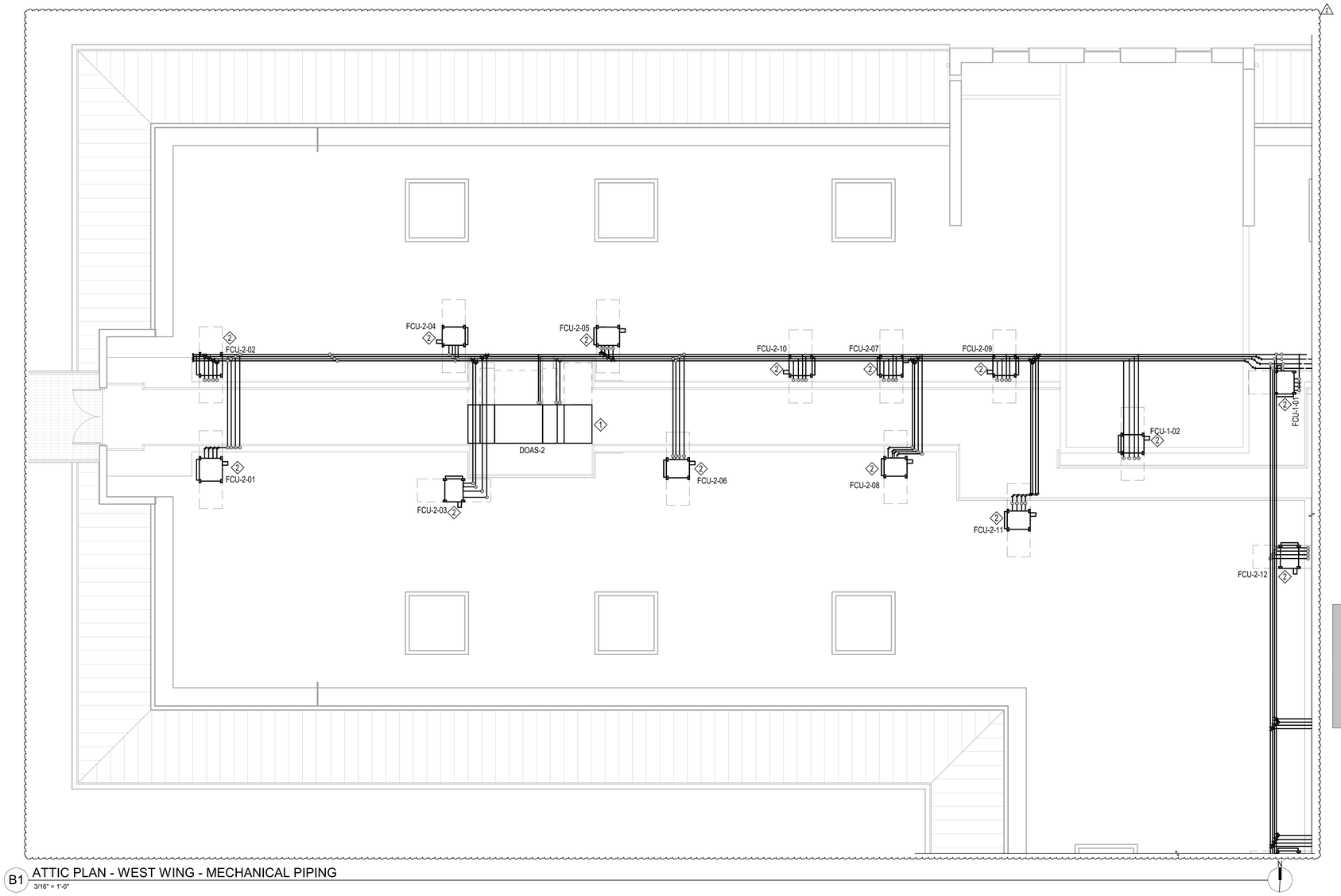
GENERAL NOTES:

1. SEE SHEET M-001 FOR GENERAL NOTES, LEGEND & ABBREVIATIONS
2. SEE CHILLED AND HOT WATER SYSTEM DIAGRAMS FOR PIPE SIZING AND MORE INFORMATION.
3. FAN COIL UNITS AND ASSOCIATED PIPING SHOWN IS LOCATED IN ATTIC AND SHOWN ON FLOOR PLANS FOR COORDINATION PURPOSES ONLY. SEE SECTION VIEWS FOR ELEVATIONS AND LOCATIONS OF EQUIPMENT.

# PLAN NOTES

1 PROVIDE DOAS MOUNTED ON EQUIPMENT PLATFORM IN ATTIC. ROUTE CHILLED AND HOT WATER PIPING TO UNIT AND PROVIDE MANUFACTURER RECOMMENDED CLEARANCES FOR SERVICE OF COILS AND VALVING.

2 PROVIDE 4-PIPE FAN COIL UNIT (FCU) ABOVE SECOND GYPSUM BOARD CEILING. UNIT TO BE INSTALLED AT ELEVATION AND ORIENTATION TO ALLOW FOR MAINTENANCE ACCESS FROM CATWALK. PROVIDE CHILLED AND HOT WATER PIPING TO 4-PIPE FAN COIL UNIT (FCU). INSTALL PER MANUFACTURER INSTALLATION INSTRUCTIONS AND PROVIDE RECOMMENDED CLEARANCES FOR SERVICE OF COILS AND VALVING. FCU TO BE PROVIDED WITH SECONDARY CONDENSATE DRAIN PAN. SEE FLOOR PLANS FOR CONTINUATION.




B1 ATTIC PLAN - WEST WING - MECHANICAL PIPING
3/16" = 1'-0"

A5 KEY PLAN - WEST WING
NTS

GRAPHIC SCALE: 3/16"=1'-0"
6' 0 4' 10'

MP108

03-09-26

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DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND

MARINE CORPS BASE

CAMP LEJEUNE, NORTH CAROLINA

DES. LWM
DR. PJR
CHK. JDL
SUBMITTED BY:
DESIGN DIR. JENNI REED
APPROVED: PWO OR OICC DATE
SATISFACTORY TO: DATE

MECHANICAL REPAIRS
BUILDING TT113

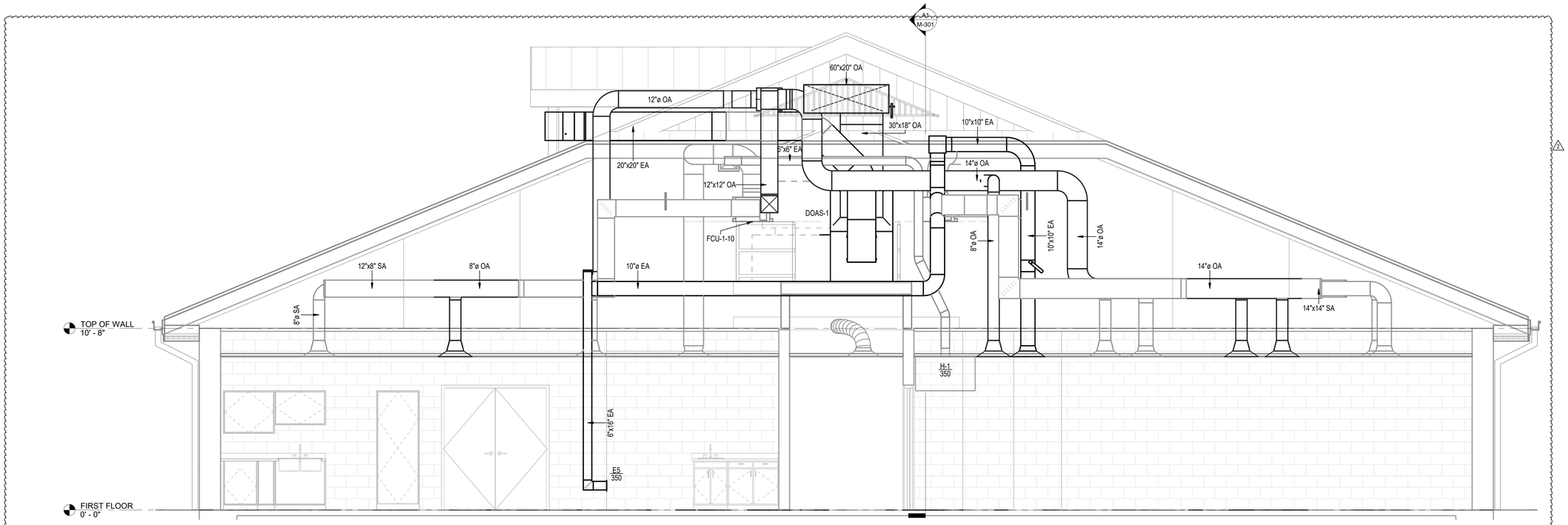
ATTIC PLAN - WEST WING - MECHANICAL PIPING

SIZE E1 | CODE IDENT. NO. 80091 | NAVFAC DRAWING NO. 60042081
CONSTR. CONTR. NO. N40085-24-F-5940
SCALE AS NOTED | SPEC. | SHEET 74 OF 119

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| 2 | PPI REVISION 2 | 03.09.26 | |
| | | | |
| | | | |

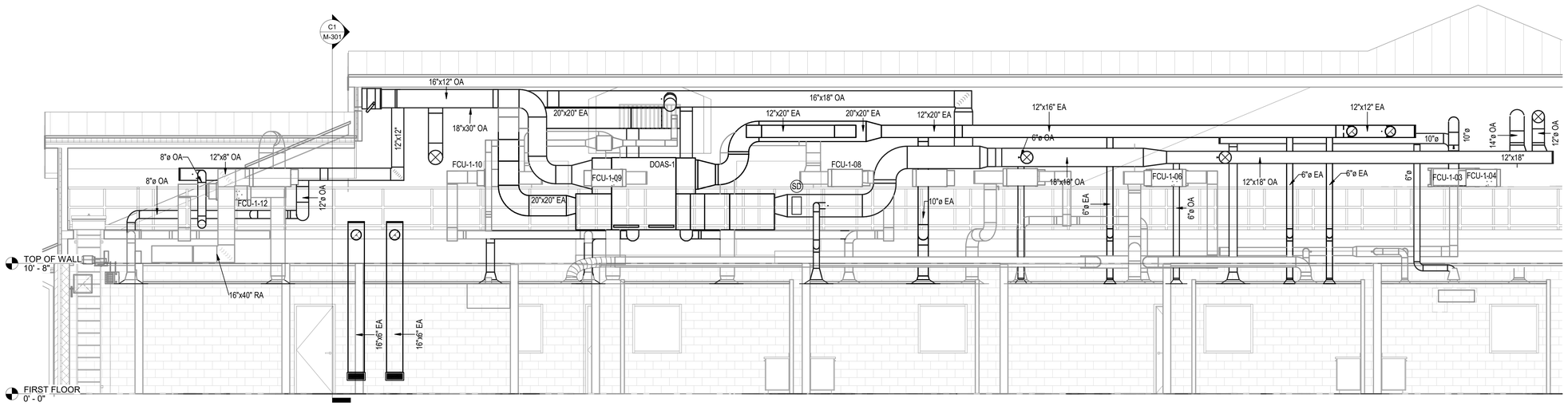
GENERAL NOTES:

1. SEE SHEET M-001 FOR GENERAL NOTES, LEGEND & ABBREVIATIONS
2. PLAN DRAWINGS ARE DIAGRAMMATIC IN NATURE AND DO NO ILLUSTRATE SPECIFIC DUCT TAKE-OFF CONFIGURATIONS OR TAPS.




C1 WEST SECTION VIEW AT DOAS-1

3/8" = 1'-0"




A1 SOUTH SECTION VIEW AT DOAS-1

1/4" = 1'-0"

GRAPHIC SCALE: 3/8"=1'-0"

3' 0 2' 6'

GRAPHIC SCALE: 1/4"=1'-0"

4' 0 2' 4' 8'

03-09-26

M-301

MBFA NO: 2406

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919-871-1070 Fax 871-5620

DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND

MARINE CORPS BASE

CAMP LEJEUNE, NORTH CAROLINA

MECHANICAL REPAIRS
BUILDING TT113

MECHANICAL SECTION VIEW

DES. LWM
DR. PJR
CHK. JDL
SUBMITTED BY:
DESIGN DIR. JENNI REED
APPROVED: PWO OR OICC DATE
SATISFACTORY TO: DATE

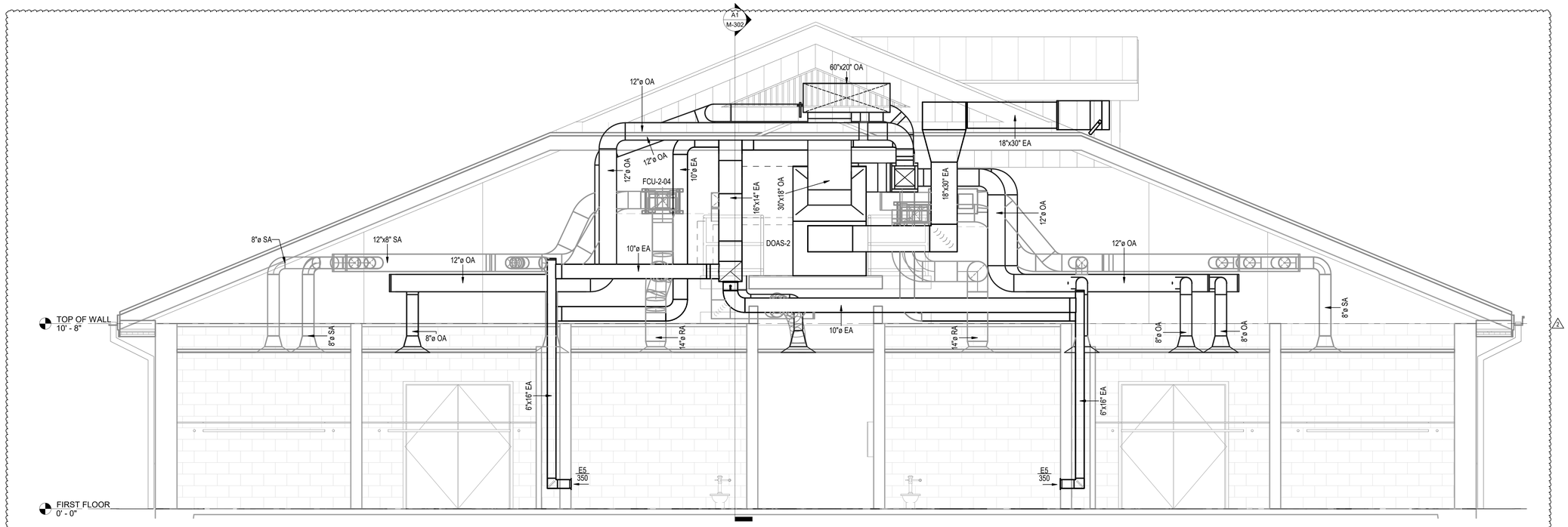
| SIZE | CODE IDENT. NO. | NAVFAC DRAWING NO. |
|---|---|---|
| E1 | 80091 | 60042083 |
| | | CONSTR. CONTR. NO. N40085-24-F-5940 |

SCALE AS NOTED SPEC. SHEET 76 OF 119

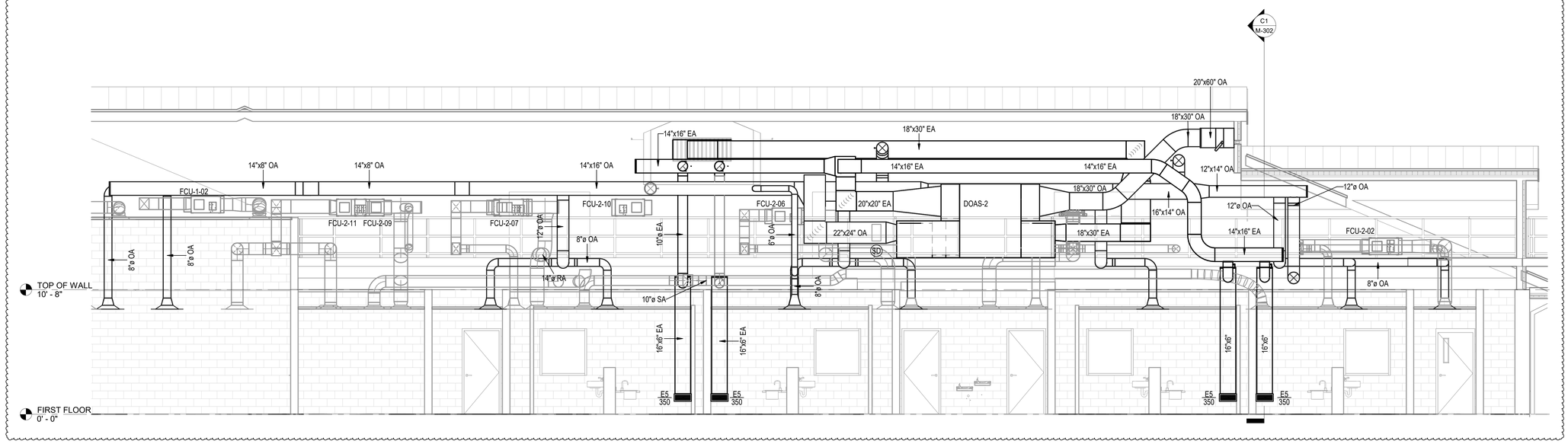
| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| 2 | PPI REVISION 2 | 03.09.26 | |

GENERAL NOTES:

1. SEE SHEET M-001 FOR GENERAL NOTES, LEGEND & ABBREVIATIONS
2. PLAN DRAWINGS ARE DIAGRAMMATIC IN NATURE AND DO NO ILLUSTRATE SPECIFIC DUCT TAKE-OFF CONFIGURATIONS OR TAPS.




C1 EAST SECTION VIEW AT DOAS-2
3/8" = 1'-0"




A1 SOUTH SECTION VIEW AT DOAS-2
1/4" = 1'-0"

GRAPHIC SCALE: 3/8"=1'-0"
3' 0 2' 6'

GRAPHIC SCALE: 1/4"=1'-0"
4' 0 2' 4' 8'

03-09-26

M-302

MBFA NO: 24016

CRENSHAW CONSULTING ENGINEERS
NC LICENSE #C-1856
www.crenshawconsulting.com
3516 Beth Street, Suite 200
Raleigh, North Carolina 27609
919-871-1070 Fax 871-5620

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MECHANICAL REPAIRS
BUILDING TT113

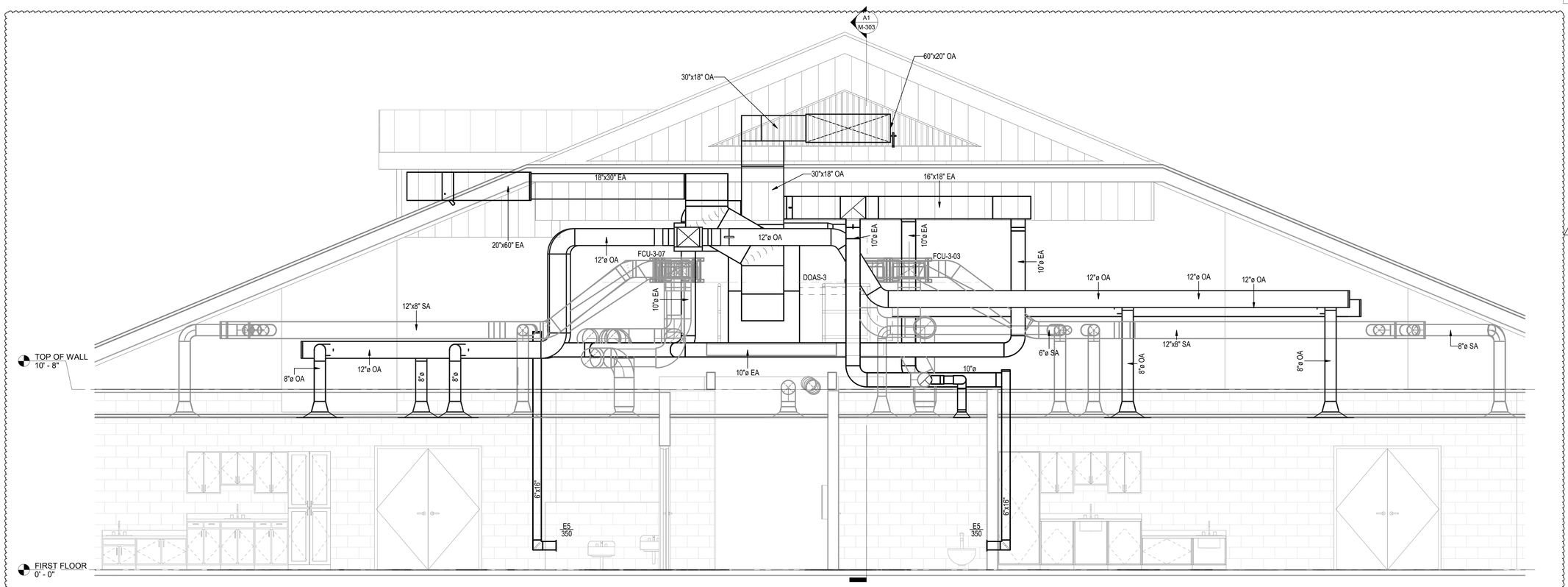
MECHANICAL SECTION VIEW

SIZE E1 | CODE IDENT. NO. 80091 | NAVFAC DRAWING NO. 60042084
CONSTR. CONTR. NO. N40085-24-F-5940
SCALE AS NOTED | SPEC. | SHEET 77 OF 119

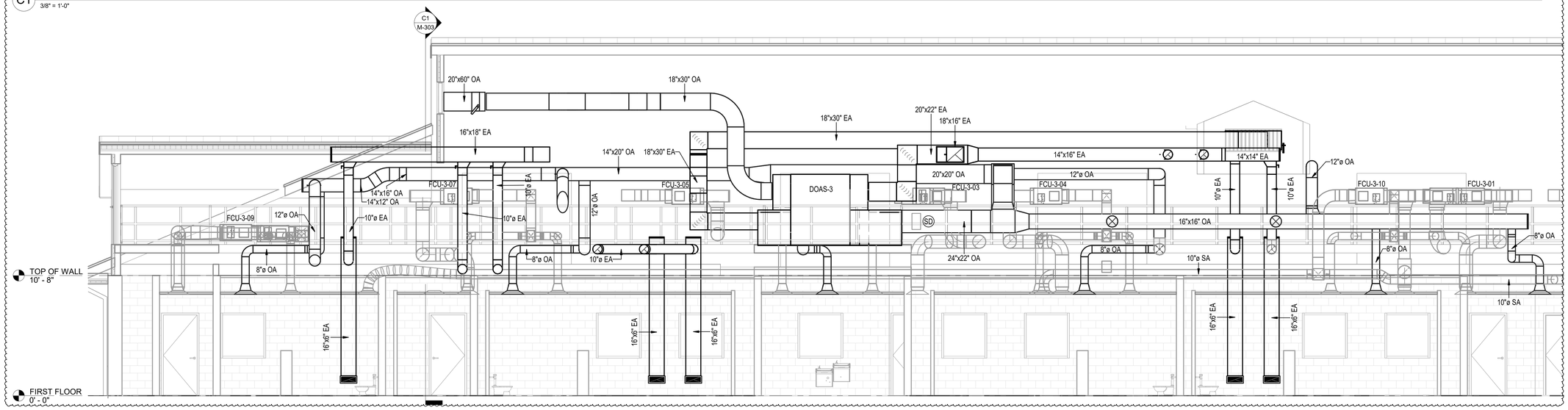
| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| 2 | PPI REVISION 2 | 03.09.26 | |

GENERAL NOTES:

1. SEE SHEET M-001 FOR GENERAL NOTES, LEGEND & ABBREVIATIONS
2. PLAN DRAWINGS ARE DIAGRAMMATIC IN NATURE AND DO NO ILLUSTRATE SPECIFIC DUCT TAKE-OFF CONFIGURATIONS OR TAPS.




C1 NORTH SECTION VIEW AT DOAS-3
3/8" = 1'-0"




A1 WEST SECTION VIEW AT DOAS-3
1/4" = 1'-0"

GRAPHIC SCALE: 3/8"=1'-0"
3' 0 2' 6'

GRAPHIC SCALE: 1/4"=1'-0"
4' 0 2' 4' 8'

M-303

MBFA NO: 2406

CRENSHAW CONSULTING engineers
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DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND

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DESIGN DIR. JENNI REED
APPROVED: PWO OR OICC DATE
SATISFACTORY TO: DATE

| SIZE | CODE IDENT. NO. | NAVFAC DRAWING NO. |
|---|---|---|
| E1 | 80091 | 60042085 |
| | | CONSTR. CONTR. NO. N40085-24-F-5940 |
| SCALE AS NOTED | SPEC. | SHEET 78 OF 119 |

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| 2 | PPI REVISION 2 | 03.09.26 | |
| | | | |

## DEDICATED OUTSIDE AIR SYSTEM (DOAS) SCHEDULE

| MARK | SUPPLY FAN SA CFM | SUPPLY FAN ESP (IN.WG) | SUPPLY FAN TSP (IN. WG) | EXHAUST FAN EA CFM | EXHAUST FAN ESP (IN. WG) | EXHAUST FAN TSP (IN. WG) | PREHEAT HEATING CAPACITY (HOT WATER) TOTAL CAP. (MBH) | EAT (°F) | LAT (°F) | FACE VEL. (FPM) | AIR PD MAX (FT. WG) | EWT (°F) | LWT (°F) | WATER PD MAX (FT. WG) | GPM |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| DOAS-1 | 3915 | 1.50 | 5.20 | 3525 | 1.50 | 3.34 | 200.0 | 22.0 | 68.0 | 550 | 5.00 | 140.0 | 110.0 | 5.00 | 13.3 |
| DOAS-2 | 3195 | 1.50 | 3.67 | 2900 | 1.50 | 2.79 | 159.2 | 22.0 | 68.0 | 550 | 5.00 | 140.0 | 110.0 | 5.00 | 10.6 |
| DOAS-3 | 3505 | 1.50 | 2.98 | 3225 | 1.50 | 2.09 | 174.6 | 22.0 | 68.0 | 550 | 5.00 | 140.0 | 110.0 | 5.00 | 11.6 |
| TOTALS: | | | | | | | 533.8 | | | | | | | | 35.6 |

| MARK | COOLING CAPACITY (CHILLED WATER) TOTAL CAP. (MBH) | SENS. CAP. (MBH) | SHR | EAT DB (°F) | EAT WB (°F) | LAT DB (°F) | LAT WB (°F) | FACE VEL. (FPM) | AIR PD MAX (FT. WG) | EWT (°F) | LWT (°F) | WATER PD MAX (FT. WG) | MIN. NUM. OF ROWS | GPM |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| DOAS-1 | 373.9 | 140.9 | 0.38 | 84.0 | 78.6 | 52.0 | 52.0 | 550 | 5.00 | 44.0 | 54.0 | 5.00 | 6 | 74.8 |
| DOAS-2 | 297.5 | 112.1 | 0.38 | 84.0 | 78.6 | 52.0 | 52.0 | 550 | 5.00 | 44.0 | 54.0 | 5.00 | 6 | 59.5 |
| DOAS-3 | 326.4 | 123.0 | 0.38 | 84.0 | 78.6 | 52.0 | 52.0 | 550 | 5.00 | 44.0 | 54.0 | 5.00 | 6 | 65.3 |
| TOTALS: | 997.8 | | | | | | | | | | | | | 199.6 |

| MARK | REHEAT HEATING CAPACITY (HOT WATER) TOTAL CAP. (MBH) | EAT (°F) | LAT (°F) | FACE VEL. (FPM) | AIR PD MAX (FT. WG) | EWT (°F) | LWT (°F) | WATER PD MAX (FT. WG) | GPM | ELECTRICAL SUPPLY FAN QTY | SUPPLY FAN BHP | SUPPLY FAN HP | EXHAUST FAN QTY | EXHAUST FAN BHP | EXHAUST FAN HP | SINGLE POINT CONNECTION FLA | SINGLE POINT CONNECTION MCA | SINGLE POINT CONNECTION MOP | V/PH | UNIT WEIGHT (LBS) | MAX DIMENSIONS LENGTH | WIDTH | HEIGHT | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| DOAS-1 | 92.5 | 55.0 | 76.0 | 550 | 5.00 | 140.0 | 110.0 | 5.00 | 6.2 | 1.0 | 5.8 | 7.5 | 1.0 | 3.6 | 5.0 | 19.9 | 22.6 | 30.0 | 480/3 | 3200 | 180" | 55" | - | 1-14 |
| DOAS-2 | 73.6 | 55.0 | 76.0 | 550 | 5.00 | 140.0 | 110.0 | 5.00 | 4.9 | 1.0 | 4.0 | 5.0 | 1.0 | 2.9 | 5.0 | 17.1 | 19.1 | 25.0 | 480/3 | 3200 | 180" | 55" | - | 1-14 |
| DOAS-3 | 80.7 | 55.0 | 76.0 | 550 | 5.00 | 140.0 | 110.0 | 5.00 | 5.4 | 1.0 | 2.9 | 5.0 | 1.0 | 4.0 | 5.0 | 17.1 | 19.1 | 25.0 | 480/3 | 3200 | 180" | 55" | - | 1-14 |
| TOTALS: | 246.8 | | | | | | | | 16.5 | | | | | | | | | | | | | | | |

REMARKS

1. PROVIDE WITH 2" MERV 8 FILTERS ON THE EXHAUST AND 2" MERV 8 PREFILTERS & 2" MERV 13 FINAL FILTERS ON THE RETURN AND OUTSIDE AIR INTAKE. STATIC PRESSURE INCREASE FROM CLEAN FILTERS IS INCLUDED IN THE SCHEDULED TOTAL STATIC PRESSURE VALUE.
2. PROVIDE PREMIUM EFFICIENCY MOTORS FOR SUPPLY AND EXHAUST FANS, COMPATIBLE WITH VARIABLE FREQUENCY DRIVES.
3. PROVIDE WITH CLASS 1A LOW LEAKAGE DAMPERS ON THE EXHAUST DISCHARGE AND OUTSIDE AIR INTAKE UNIT CONNECTIONS.
4. PROVIDE UNIT WITH HORIZONTAL SUPPLY DISCHARGE, EXHAUST INLET AND OUTSIDE AIR INLET DUCT CONNECTION. UNIT MUST HAVE 2" DOUBLE WALL CONSTRUCTION.
5. VERIFY THAT UNIT CAN BE INSTALLED IN LOCATION SHOWN ON DRAWINGS PRIOR TO SUBMITTING FOR APPROVAL.
6. PROVIDE A 10,000 HR SALT SPRAY PROTECTIVE COATING ON THE COOLING AND REHEAT COILS.
7. UNIT MUST HAVE CROSSFLOW FIXED PLATE TOTAL ENTHALPY HEAT EXCHANGER, PLENUM SUPPLY/EXHAUST FANS, COOLING COIL, REHEAT COIL AND MARINE LIGHTS IN EACH FAN & ACCESS SECTIONS.
8. PROVIDE UNIT WITH SINGLE POINT CONNECTION.
9. REFER TO CV SCHEDULE FOR CONTROL VALVE REQUIREMENTS.
10. PROVIDE UNIT WITH RECIRCULATION PUMP AT PREHEAT COIL FOR FREEZE PROTECTION. ALSO PROVIDE 120/1 CIRCUIT FOR PUMP. PUMP IS SIZED FOR 110% OF PREHEAT COIL PRESSURE DROP AND DESIGN FLOW GPM. SEE HOT WATER PREHEAT COIL DETAIL AND CONTROLS FOR FURTHER INFORMATION.
11. MAXIUMUM ALLOWABLE DISCHARGE OR RADIATED NOISE CRITERIA (NC) = 30.
12. AIR HANDLER DESIGNED WITH A FOULING FACTOR OF 0.00025 FT^2-HR-°F/BTU FOR ALL COILS.
13. PROVIDE UNIT WITH A MINIMUM 8" INTEGRAL BASE FRAME. EQUIPMENT SUPPORTS TO HAVE VIBRATION ISOLATION. SEE DETAILS AND STRUCTURAL DRAWINGS FOR MORE INFORMATION.
14. PROVIDE VENTILATION SUPPLY DUCTWORK WITH SMOKE DETECTOR.

### TOTAL ENERGY HEAT EXCHANGER

| SUMMER OA DB | SUMMER OA WB | SUMMER OA DB | SUMMER OA %RH | SUMMER RA DB | SUMMER RA WB | WINTER OA DB | WINTER OA WB | WINTER OA DB | WINTER OA %RH | WINTER RA DB | WINTER RA WB | HX EFF. SUMMER/ WINTER % |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 83.3 | 78.5 | 76.0 | 50.0 | 81.2 | 77.0 | 26.8 | 22.4 | 70.0 | 35.0 | 48.8 | 35.9 | 35.4 / 63.7 |
| 83.3 | 78.5 | 76.0 | 50.0 | 81.7 | 76.9 | 26.8 | 22.4 | 70.0 | 35.0 | 48.1 | 35.5 | 39 / 65.3 |
| 83.3 | 78.5 | 76.0 | 50.0 | 80.1 | 76.9 | 26.8 | 22.4 | 70.0 | 35.0 | 51.7 | 37.5 | 58.8 / 76.7 |

## DESIGN CONDITIONS

### OUTDOOR DESIGN CONDITIONS

| SEASON | DB/WB (°F) |
|---|---|
| SUMMER | 91/77 (1% ASHRAE) |
| WINTER | 22 (99.6% ASHRAE) |

### DEHUMIDIFICATION CONDITIONS

| MCDB (°F) | W (grains H2O/lbm dry air) |
|---|---|
| 83.3 (1.0% ASHRAE) | 140.62 (1.0% ASHRAE) |

### INDOOR DESIGN CONDITIONS

| SPACE CLASSIFICATION | SEASON | OCCUPIED | UNOCCUPIED | ALARM | RELATIVE HUMIDITY |
|---|---|---|---|---|---|
| FIRE ALARM/SPRINKLER | COOLING | 120°F (MAX) | - | 120°F (MAX) | 0-85% |
| | HEATING | 32°F (MIN) | - | 32°F (MIN) | 0-85% |
| NFPA 72 10.3.5. FOLLOW THIS OR MANUFACTURER'S REQUIREMENTS, WHICHEVER IS MORE STRINGENT. | | | | | |
| ELECTRICAL ROOM | COOLING | 95°F (MAX) | - | 95°F (MAX) | 80% (MAX) |
| | HEATING | 41°F (MIN) | - | 41°F (MIN) | 8% (MIN) |
| EXTERIOR OR PRIMARY ELECTRICAL EQUIPMENT ROOMS ONLY. OTHERWISE FOLLOW UFC 3-410-01. | | | | | |
| TELECOM ROOM | COOLING | 68°F | - | 95°F (MAX) | 80% (MAX) |
| | HEATING | - | - | 41°F (MIN) | 8% (MIN) |
| ALL OTHER ROOMS | COOLING | 75°F | 80°F | 85°F | 50% |
| | HEATING | 70°F | 60°F | 55°F | 50% |

## DUCT-MOUNTED MOTORIZED DAMPER (MD) SCHEDULE

| MARK | SERVES | FLOW | SIZE WxH (in.) | VOLT./PH | REMARKS |
|---|---|---|---|---|---|
| MD-1 | EF-1 | INTAKE | 18X18 | 120/1 | 1 |
| MD-2 | EF-2 | INTAKE | 18X18 | 120/1 | 1 |
| MD-3 | EF-3 | INTAKE | 18X18 | 120/1 | 1 |
| MD-4 | EF-4 | INTAKE | 36X36 | 120/1 | 1 |
| MD-5 | HOOD | EXHAUST | 10X10 | 120/1 | 1 |
| MD-6 | DOAS-1, EF-5 | EXHAUST | 60X20 | 120/1 | 1 |
| MD-7 | DOAS-1 | INTAKE | 60X20 | 120/1 | 1 |
| MD-8 | DOAS-2 | EXHAUST | 60X20 | 120/1 | 1 |
| MD-9 | DOAS-2 | INTAKE | 60X20 | 120/1 | 1 |
| MD-10 | DOAS-3 | EXHAUST | 60X20 | 120/1 | 1 |
| MD-11 | DOAS-3 | INTAKE | 60X20 | 120/1 | 1 |

REMARKS :

1. CLASS 1A LOW LEAKAGE MOTORIZED DAMPER.

## AIR DOOR (AD) SCHEDULE

| MARK | AREA SERVED | MANUFACTURER | MODEL NO. | LENGTH | CFM | MOTOR HP | FLA | MOCP | VOLT/PH | WEIGHT | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|
| AD-1 | KITCHEN DOOR ENTRY | MARS | STD242 | 42" | 1418 | 0.5 | 5.1 | 15 | 120/1 | 70 | 1,2 |

REMARKS:

1. PROVIDE WITH MOTOR CONTROL PANEL AND WALL MOUNTING BRACKET.
2. PROVIDE WITH DOOR POSITION SWITCH OR MAGNETIC REED SWITCH, INTERLOCKED TO OPERATE IF DOOR IS OPEN.

## AIR DISTRIBUTION SCHEDULE

| MARK | DESCRIPTION | THROW | FACE SIZE | NECK SIZE | MOUNT | MINIMUM CFM | MAXIMUM CFM | MAX. NC | REMARKS |
|---|---|---|---|---|---|---|---|---|---|
| S1 | SQUARE CONE DIFFUSER | 4 WAY | 12X12 | 6"ø | LAY-IN | 0 CFM | 75 CFM | 30 | 1-4 |
| S2 | SQUARE CONE DIFFUSER | 4 WAY | 24x24 | 8"ø | LAY-IN | 100 CFM | 200 CFM | 30 | 1-4 |
| S3 | SQUARE CONE DIFFUSER | 4 WAY | 24x24 | 10"ø | LAY-IN | 225 CFM | 275 CFM | 30 | 1-4 |
| S4 | SQUARE CONE DIFFUSER | 4 WAY | 24x24 | 12"ø | LAY-IN | 300 CFM | 600 CFM | 30 | 1-4 |
| | | | | | | | | | |
| E1 | PERFORATED EXHAUST DIFFUSER | NA | 12X12 | 6"ø | LAY-IN | 0 CFM | 75 CFM | 30 | 1-4 |
| E2 | PERFORATED EXHAUST DIFFUSER | NA | 24x24 | 8"ø | LAY-IN | 100 CFM | 175 CFM | 30 | 1-4 |
| E3 | PERFORATED EXHAUST DIFFUSER | NA | 24x24 | 10"ø | LAY-IN | 200 CFM | 275 CFM | 30 | 1-4 |
| E4 | PERFORATED EXHAUST DIFFUSER | NA | 24x24 | 12"ø | LAY-IN | 300 CFM | 400 CFM | 30 | 1-4 |
| E5 | LOUVERED, 3/4" SPACING GRILLE | NA | 16X12 | 16X12 | SURFACE | 250 CFM | 375 CFM | 30 | |
| | | | | | | | | | |
| R1 | PERFORATED RETURN DIFFUSER | NA | 12X12 | 6"ø | LAY-IN | 0 CFM | 75 CFM | 30 | 1-4 |
| R2 | PERFORATED RETURN DIFFUSER | NA | 24X24 | 8"ø | LAY-IN | 100 CFM | 175 CFM | 30 | 1-4 |
| R3 | PERFORATED RETURN DIFFUSER | NA | 24X24 | 10"ø | LAY-IN | 200 CFM | 275 CFM | 30 | 1-4 |
| R4 | PERFORATED RETURN DIFFUSER | NA | 24X24 | 12"ø | LAY-IN | 300 CFM | 400 CFM | 30 | 1-4 |
| R5 | PERFORATED RETURN DIFFUSER | NA | 24X24 | 14"ø | LAY-IN | 425 CFM | 700 CFM | 30 | 1-4 |
| R6 | PERFORATED RETURN DIFFUSER | NA | 24X24 | 16"ø | LAY-IN | 725 CFM | 950 CFM | 30 | 1-4 |
| R7 | PERFORATED RETURN DIFFUSER | NA | 24X24 | 16"ø | SURFACE | 940 | 940 | 30 | |
| | | | | | | | | | |

REMARKS

1. VERIFY CONSTRUCTION TYPES WITH ARCHITECTURAL PLANS TO DETERMINE MOUNTING DETAILS AND ACCESSORIES REQUIRED. COORDINATE COLOR W/ ARCHITECT.
2. PROVIDE WITH SQUARE TO ROUND TRANSITION.
3. ALL AIR DISTRIBUTION MUST BE 100% ALUMINUM CONSTRUCTION.
4. PROVIDE BLANKET INSULATION ON THE BACK OF ALL DIFFUSERS.

## FAN COIL UNIT (FCU) SCHEDULE

| MARK | COOLING CAPACITY (CHW) MAX CFM | MIN CFM | E.S.P. "WC | T.S.P. "WC | TOTAL MBH | SENSIBLE MBH | EAT (DB) °F | EAT (WB) °F | LAT (DB) °F | LAT (WB) °F | WATER TEMP. (°F) EWT/LWT | WATER PD FT. MAX. | VELOCITY FPS. MAX | MIN. NUM. OF ROWS | GPM |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FCU-1-01 | 300 | 250 | 0.50 | 0.86 | 9.1 | 7.0 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 1.8 |
| FCU-1-02 | 500 | 250 | 0.50 | 1.15 | 12.1 | 9.3 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 2.4 |
| FCU-1-03 | 400 | 400 | 0.50 | 1.15 | 12.1 | 9.3 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 2.4 |
| FCU-1-04 | 500 | 375 | 0.50 | 1.44 | 15.1 | 11.6 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 3.0 |
| FCU-1-05 | 400 | 250 | 0.50 | 1.15 | 12.1 | 9.3 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 2.4 |
| FCU-1-06 | 500 | 375 | 0.50 | 1.44 | 15.1 | 11.6 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 3.0 |
| FCU-1-07 | 500 | 375 | 0.50 | 1.44 | 15.1 | 11.6 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 3.0 |
| FCU-1-08 | 500 | 375 | 0.50 | 1.44 | 15.1 | 11.6 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 3.0 |
| FCU-1-09 | 500 | 375 | 0.50 | 1.44 | 15.1 | 11.6 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 3.0 |
| FCU-1-10 | 500 | 375 | 0.50 | 1.44 | 15.1 | 11.6 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 3.0 |
| FCU-1-11 | 1375 | 800 | 0.50 | 1.05 | 41.6 | 32.0 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 8.3 |
| FCU-1-12 | 600 | 375 | 0.50 | 1.42 | 18.1 | 14.0 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 3.6 |
| FCU-1-13 | 500 | 375 | 0.50 | 1.44 | 15.1 | 11.6 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 3.0 |
| FCU-1-14 | 525 | 250 | 0.50 | 1.44 | 15.1 | 11.6 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 3.0 |
| FCU-2-01 | 600 | 375 | 0.50 | 1.42 | 18.1 | 14.0 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 3.6 |
| FCU-2-02 | 600 | 375 | 0.50 | 1.42 | 18.1 | 14.0 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 3.6 |
| FCU-2-03 | 500 | 375 | 0.50 | 1.44 | 15.1 | 11.6 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 3.0 |
| FCU-2-04 | 500 | 375 | 0.50 | 1.44 | 15.1 | 11.6 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 3.0 |
| FCU-2-05 | 500 | 375 | 0.50 | 1.44 | 15.1 | 11.6 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 3.0 |
| FCU-2-06 | 500 | 375 | 0.50 | 1.44 | 15.1 | 11.6 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 3.0 |
| FCU-2-07 | 500 | 375 | 0.50 | 1.44 | 15.1 | 11.6 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 3.0 |
| FCU-2-08 | 500 | 375 | 0.50 | 1.44 | 15.1 | 11.6 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 3.0 |
| FCU-2-09 | 300 | 250 | 0.50 | 0.86 | 9.1 | 7.0 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 1.8 |
| FCU-2-10 | 500 | 375 | 0.50 | 1.44 | 15.1 | 11.6 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 3.0 |
| FCU-2-11 | 750 | 500 | 0.50 | 1.23 | 23.9 | 17.5 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 4.8 |
| FCU-2-12 | 500 | 250 | 0.50 | 1.15 | 12.1 | 9.3 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 2.4 |
| FCU-3-01 | 700 | 375 | 0.50 | 1.15 | 21.1 | 16.3 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 4.2 |
| FCU-3-02 | 600 | 375 | 0.50 | 1.42 | 18.1 | 14.0 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 3.6 |
| FCU-3-03 | 700 | 500 | 0.50 | 1.15 | 21.1 | 16.3 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 4.2 |
| FCU-3-04 | 600 | 375 | 0.50 | 1.42 | 18.1 | 14.0 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 3.6 |
| FCU-3-05 | 700 | 500 | 0.50 | 1.15 | 21.1 | 16.3 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 4.2 |
| FCU-3-06 | 600 | 375 | 0.50 | 1.42 | 18.1 | 14.0 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 3.6 |
| FCU-3-07 | 600 | 375 | 0.50 | 1.42 | 18.1 | 14.0 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 3.6 |
| FCU-3-08 | 800 | 500 | 0.50 | 1.31 | 24.2 | 18.6 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 4.8 |
| FCU-3-09 | 700 | 500 | 0.50 | 1.15 | 21.1 | 16.3 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 4.2 |
| FCU-3-10 | 650 | 375 | 0.50 | 1.07 | 19.7 | 15.1 | 76.2 | 67.0 | 57.0 | 57.0 | 44 / 54 | 5.0 | 3 | 4 | 3.9 |
| **TOTALS:** | | | | | **613.6** | | | | | | | | | | **122.7** |

| MARK | HEATING CAPACITY (HW) HEATING TOTAL MBH | HEATING EAT/LAT °F | HEATING EWT/LWT °F | HEATING GPM | REHEAT (DEHUMIDIFICATION) TOTAL MBH | REHEAT EAT/LAT °F | REHEAT EWT/LWT °F | REHEAT GPM | WATER PD FT. MAX | VELOCITY FPS, MAX | ROWS | ELECTRICAL V/PH | BHP | ECM HP | MCA/MOP | UNIT WEIGHT (LBS) | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FCU-1-01 | 8.1 | 65 / 95 | 140 / 110 | 0.70 | 4.6 | 55 / 72 | 140 / 110 | 0.35 | 5.0 | 4 | 2 | 277/1 | 0.085 | 0.645 | 4.7 / 15 | 120 | 1,2,3,4,5,6 |
| FCU-1-02 | 8.1 | 65 / 95 | 140 / 110 | 0.70 | 4.6 | 55 / 72 | 140 / 110 | 0.35 | 5.0 | 4 | 2 | 277/1 | 0.145 | 0.693 | 4.7 / 15 | 120 | 1,2,3,4,5,6 |
| FCU-1-03 | 13.0 | 65 / 95 | 140 / 110 | 1.05 | 7.3 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.145 | 0.693 | 4.7 / 15 | 120 | 1,2,3,4,5,6 |
| FCU-1-04 | 12.2 | 65 / 95 | 140 / 110 | 1.05 | 6.9 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.242 | 0.723 | 4.7 / 15 | 120 | 1,2,3,4,5,6 |
| FCU-1-05 | 8.1 | 65 / 95 | 140 / 110 | 0.70 | 4.6 | 55 / 72 | 140 / 110 | 0.35 | 5.0 | 4 | 2 | 277/1 | 0.145 | 0.693 | 4.7 / 15 | 120 | 1,2,3,4,5,6 |
| FCU-1-06 | 12.2 | 65 / 95 | 140 / 110 | 1.05 | 6.9 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.242 | 0.723 | 4.7 / 15 | 120 | 1,2,3,4,5,6 |
| FCU-1-07 | 12.2 | 65 / 95 | 140 / 110 | 1.05 | 6.9 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.242 | 0.723 | 4.7 / 15 | 120 | 1,2,3,4,5,6 |
| FCU-1-08 | 12.2 | 65 / 95 | 140 / 110 | 1.05 | 6.9 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.242 | 0.723 | 4.7 / 15 | 120 | 1,2,3,4,5,6 |
| FCU-1-09 | 12.2 | 65 / 95 | 140 / 110 | 1.05 | 6.9 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.242 | 0.723 | 4.7 / 15 | 120 | 1,2,3,4,5,6 |
| FCU-1-10 | 12.2 | 65 / 95 | 140 / 110 | 1.05 | 6.9 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.242 | 0.723 | 4.7 / 15 | 120 | 1,2,3,4,5,6 |
| FCU-1-11 | 25.9 | 65 / 95 | 140 / 110 | 1.75 | 14.7 | 55 / 72 | 140 / 110 | 1.05 | 5.0 | 4 | 2 | 277/1 | 0.521 | 1.198 | 7.23 / 15 | 120 | 1,2,3,4,5,6 |
| FCU-1-12 | 12.2 | 65 / 95 | 140 / 110 | 1.05 | 6.9 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.355 | 0.732 | 4.7 / 15 | 120 | 1,2,3,4,5,6 |
| FCU-1-13 | 12.2 | 65 / 95 | 140 / 110 | 1.05 | 6.9 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.242 | 0.723 | 4.7 / 15 | 120 | 1,2,3,4,5,6 |
| FCU-1-14 | 8.1 | 65 / 95 | 140 / 110 | 0.70 | 4.6 | 55 / 72 | 140 / 110 | 0.35 | 5.0 | 4 | 2 | 277/1 | 0.242 | 0.723 | 4.7 / 15 | 120 | 1,2,3,4,5,6 |
| FCU-2-01 | 12.2 | 65 / 95 | 140 / 110 | 1.05 | 6.9 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.355 | 0.732 | 4.7 / 15 | 120 | 1,2,3,4,5,6 |
| FCU-2-02 | 12.2 | 65 / 95 | 140 / 110 | 1.05 | 6.9 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.355 | 0.732 | 4.7 / 15 | 120 | 1,2,3,4,5,6 |
| FCU-2-03 | 12.2 | 65 / 95 | 140 / 110 | 1.05 | 6.9 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.242 | 0.723 | 4.7 / 15 | 120 | 1,2,3,4,5,6 |
| FCU-2-04 | 12.2 | 65 / 95 | 140 / 110 | 1.05 | 6.9 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.242 | 0.723 | 4.7 / 15 | 120 | 1,2,3,4,5,6 |
| FCU-2-05 | 12.2 | 65 / 95 | 140 / 110 | 1.05 | 6.9 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.242 | 0.723 | 4.7 / 15 | 120 | 1,2,3,4,5,6 |
| FCU-2-06 | 12.2 | 65 / 95 | 140 / 110 | 1.05 | 6.9 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.242 | 0.723 | 4.7 / 15 | 120 | 1,2,3,4,5,6 |
| FCU-2-07 | 12.2 | 65 / 95 | 140 / 110 | 1.05 | 6.9 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.242 | 0.723 | 4.7 / 15 | 120 | 1,2,3,4,5,6 |
| FCU-2-08 | 12.2 | 65 / 95 | 140 / 110 | 1.05 | 6.9 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.242 | 0.723 | 4.7 / 15 | 120 | 1,2,3,4,5,6 |
| FCU-2-09 | 8.1 | 65 / 95 | 140 / 110 | 0.70 | 4.6 | 55 / 72 | 140 / 110 | 0.35 | 5.0 | 4 | 2 | 277/1 | 0.085 | 0.645 | 4.7 / 15 | 120 | 1,2,3,4,5,6 |
| FCU-2-10 | 12.2 | 65 / 95 | 140 / 110 | 1.05 | 6.9 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.242 | 0.723 | 4.7 / 15 | 120 | 1,2,3,4,5,6 |
| FCU-2-11 | 16.2 | 65 / 95 | 140 / 110 | 1.40 | 9.2 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.517 | 0.733 | 4.7 / 15 | 130 | 1,2,3,4,5,6 |
| FCU-2-12 | 8.1 | 65 / 95 | 140 / 110 | 0.70 | 4.6 | 55 / 72 | 140 / 110 | 0.35 | 5.0 | 4 | 2 | 277/1 | 0.145 | 0.693 | 4.7 / 15 | 120 | 1,2,3,4,5,6 |
| FCU-3-01 | 12.2 | 65 / 95 | 140 / 110 | 1.05 | 6.9 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.145 | 0.693 | 4.7 / 15 | 130 | 1,2,3,4,5,6 |
| FCU-3-02 | 12.2 | 65 / 95 | 140 / 110 | 1.05 | 6.9 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.355 | 0.732 | 4.7 / 15 | 130 | 1,2,3,4,5,6 |
| FCU-3-03 | 16.2 | 65 / 95 | 140 / 110 | 1.40 | 9.2 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.145 | 0.693 | 4.7 / 15 | 130 | 1,2,3,4,5,6 |
| FCU-3-04 | 12.2 | 65 / 95 | 140 / 110 | 1.05 | 6.9 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.355 | 0.732 | 4.7 / 15 | 130 | 1,2,3,4,5,6 |
| FCU-3-05 | 16.2 | 65 / 95 | 140 / 110 | 1.40 | 9.2 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.145 | 0.693 | 4.7 / 15 | 130 | 1,2,3,4,5,6 |
| FCU-3-06 | 12.2 | 65 / 95 | 140 / 110 | 1.05 | 6.9 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.355 | 0.732 | 4.7 / 15 | 130 | 1,2,3,4,5,6 |
| FCU-3-07 | 12.2 | 65 / 95 | 140 / 110 | 1.05 | 6.9 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.355 | 0.732 | 4.7 / 15 | 130 | 1,2,3,4,5,6 |
| FCU-3-08 | 16.2 | 65 / 95 | 140 / 110 | 1.40 | 9.2 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.609 | 0.733 | 4.7 / 15 | 130 | 1,2,3,4,5,6 |
| FCU-3-09 | 16.2 | 65 / 95 | 140 / 110 | 1.40 | 9.2 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.145 | 0.693 | 4.7 / 15 | 130 | 1,2,3,4,5,6 |
| FCU-3-10 | 12.2 | 65 / 95 | 140 / 110 | 1.05 | 6.9 | 55 / 72 | 140 / 110 | 0.70 | 5.0 | 4 | 2 | 277/1 | 0.36 | 0.725 | 4.7 / 15 | 130 | 1,2,3,4,5,6 |
| **TOTALS:** | | | | **38.1** | | | | **23.4** | | | | | | | | | |

REMARKS:

1. PROVIDE WITH TOP ACCESS FILTER SECTION WITH 2" MERV-8 EFFICIENCY THROWAWAY FLAT FILTER. PROVIDE WITH FIVE (5) SETS TOTAL: ONE (1) FOR STARTUP, ONE (1) FOR BOD, TWO (2) IN BOX, AND ONE (1) FOR 2ND SEASON. CHANGE FILTERS AT BOD AND 2ND SEASON TAB.
2. PROVIDE ECM MOTOR FOR SUPPLY FAN. UNIT LEFT HAND AND RIGHT HAND CONFIGURATION FOR CONTROL PANEL AND COIL CONNECTIONS MUST BE PROVIDED TO ALLOW FOR ACCESS FROM CATWALK.
3. PROVIDE UNIT WITH HORIZONTAL SUPPLY DISCHARGE AND HORIZONTAL RETURN INLET DUCT CONNECTIONS. UNIT MUST HAVE 2" DOUBLE WALL CONSTRUCTION.
4. VERIFY THAT UNIT CAN BE INSTALLED IN LOCATION SHOWN ON DRAWINGS PRIOR TO SUBMITTING FOR APPROVAL.
5. PROVIDE UNIT WITH SINGLE POINT POWER CONNECTION.
6. PROVIDE UNIT WITH SECONDARY DRAIN PAN AND DRAIN PAN FLOAT SWITCH TO CLOSE CHILLED WATER VALVE UPON ALARM.

2

CARL W. MOORE JR. 056500 REGISTERED P

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| 2 | PPI REVISION 2 | 03.09.26 | |
| | | | |

## EXPANSION TANK (ET) SCHEDULE

| MARK | FLUID | PIPING MATERIAL | SYSTEM VOLUME (GAL) | MIN. OPERATING TEMP. (F) | MAX. DESIGN TEMP. (F) | MAX. SYSTEM HEIGHT (FT.) | PRECHARGE PRESS. (PSI) | ASME PRESSURE RATING (PSI) | TANK VOLUME (GAL) | ACCEPTANCE VOLUME (GAL) | SERVICE | WEIGHT (LB) | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | EMPTY | FULL | |
| ET-1 | WATER | COPPER | 475 | 50 | 140 | 21 | 18 | 125 | 10 | 10 | HOT WATER | 40 | 124 | 1 |
| ET-2 | WATER | COPPER | 2258 | 42 | 90 | 28 | 18 | 125 | 13 | 13 | CHILLED WATER | 120 | 250 | 1 |
| | | | | | | | | | | | | | | |

REMARKS:
1. PROVIDE PRE-CHARGED FULL ACCEPTANCE BLADDER TANK AS WELL AS SIGHT GLASS, ANCHOR CLIPS, AND STAINLESS STEEL CONNECTOR.

## BUFFER TANK (BT) SCHEDULE

| MARK | GALLONS | DIMENSIONS | CONNECTION SIZE (IN) | WEIGHT (LBS) | SERVICE | REMARKS |
|---|---|---|---|---|---|---|
| BT-1 | 1500 | 5' DIA X 12' 4" LONG | 6" | 15250 | CHILLED WATER | 1 |
| | | | | | | |

REMARKS
1. PROVIDE WITH INTERNAL BAFFLE, AIR VENT AT TOP OF TANK, DRAIN, FLANGED LOW CONNECTIONS AND
4 LEG STANDS FOR MOUNTING TO CONCRETE PAD.

## AIR AND DIRT SEPERATOR (AS) SCHEDULE

| MARK | CONNECTION SIZE (IN) | MAX PD (FT. WG) | SERVICE | REMARKS |
|---|---|---|---|---|
| AS-1 | 3 | 5 | HOT WATER | 1,2 |
| AS-2 | 4 | 5 | CHILLED WATER | 1,2 |
| | | | | |

REMARKS
1. PROVIDE WITH REMOVABLE HEAD AND AUTOMATIC AIR RELIEF VALVE.
2. SEPARATOR MUST BE COALESCING AIR/DIRT TYPE.

## HOT WATER UNIT HEATER (UH) SCHEDULE

| MARK | LOCATION | THROW | AIRFLOW (CFM) | HEATING CAPACITY (MBH) | EWT | LWT | FLOW (GPM) | PD (FT. HD) | FAN (HP) | VOLT/PH | WEIGHT | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| UH-1 | 109F MECH. BLDG. | HORIZONTAL | 340 | 18.0 | 140 | 110 | 1.2 | 0.2 | 1/60 | 115/1 | 16 | 1,2,3 |
| UH-2 | 139 MECHANICAL | HORIZONTAL | 340 | 18.0 | 140 | 110 | 1.2 | 0.2 | 1/60 | 115/1 | 16 | 1,2,3 |
| UH-3 | 141 SPRINKLER | HORIZONTAL | 340 | 18.0 | 140 | 110 | 1.2 | 0.2 | 1/60 | 115/1 | 16 | 1,2,3 |
| | | | | | | | | | | | | |

REMARKS
1. PROVIDE WITH WALL/CEILING MOUNTING BRACKET.
2. PROVIDE WITH WALL MOUNTED, LINE VOLTAGE THERMOSTAT. E.C. TO INSTALL AND WIRE.
3. SEE CONTROL VALVE SCHEDULE FOR CONTROL VALVE REQUIREMENTS.

## CHEMICAL SHOT FEEDER (CF) SCHEDULE

| MARK | TANK VOLUME (GAL) | SERVICE | WEIGHT (LB) | | REMARKS |
|---|---|---|---|---|---|
| | | | EMPTY | FULL | |
| CF-1 | 2 | HOT WATER | 20 | 40 | 1 |
| CF-2 | 5 | CHILLED WATER | 40 | 80 | 1 |
| | | | | | |

REMARKS:
1. PROVIDE 200 PSIG PRESSURE RATED POT FEEDER.

## AIR COOLED PACKAGE CHILLER (CH) SCHEDULE

| MARK | NOMINAL CAPACITY (TONS) | COOLING CAPACITY (TONS) | EWT (°F) | LWT (°F) | Δ P (FT) | GPM | FOULING FACTOR ( °F FT2 H/BTU) | FULL LOAD EER | IPLV EER | VOLT/PH | MCA | MOCP | WEIGHT (LBS) | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CH-1,2 | 90 | 85.9 | 54 | 44 | 12 | 213 | 0.0001 | 10.7 | 16.8 | 460/3 | 183 | 225 | 6000 | 1,2,3,4,5 |
| | | | | | | | | | | | | | | |

REMARKS
1. PROVIDE MINIMUM 4 SCROLL COMPRESSORS, LOW AMBIENT OPERATION DOWN TO 0°F, HAIL GUARDS ON COIL AND COMPRESSOR SECTIONS, VIBRATION ISOLATION, FLOW SWITCH,
BACNET MS/TP COMMUNICATIONS CARD FOR FULL INTERGRATION OF POINTS INTO BAS AND SINGLE POINT POWER CONNECTION.
2. SELECT UNIT AT 100° AMBIENT WITH R-454B OR R-32. CHILLER MUST HAVE TWO INDEPENDENT REFRIGERATION CIRCUITS.
3. PROVIDE FACTORY START-UP, 1 YEAR LABOR WARRANTY, 3 YEAR CONDENSER COIL WARRANTY AND 10 YEAR EXTENDED COMPRESSOR WARRANTY.
4. ENTIRE CHILLER MUST HAVE 10,000 HR SALTSPRAY PROTECTION IN ACCORDANCE WITH ASTM B117. COATINGS MUST NOT REDUCE COIL CAPACITIES BELOW SCHEDULED VALUES.
5. PROVIDE CORE FILTER DRYER ON SUCTION LINE AND LIQUID AND SUCTION LINE SERVICE VALVES.

## BOILER (B) SCHEDULE

| MARK | LOCATION | ORIENTATION | INPUT (MBH) | OUTPUT (MBH) | TURNDOWN | AFUE MIN EFF % | EWT (°F) | LWT (°F) | FLOW (GPM) | PD (FT. HD) | FLUE SIZE (IN) | INTAKE SIZE (IN) | ELECTRICAL | | | WEIGHT (LBS) | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | VOLT/PH | FLA | MCA | | |
| B-1 | MECH ROOM | CONDENSING FLOOR MOUNTED | 399 | 387 | 10:1 | 97.0 | 110 | 140 | 33 | 8 | 4 | 4 | 120/1 | 3.0 | 3.8 | 400 | 1,2,3,4,5,6 |
| B-2 | MECH ROOM | CONDENSING FLOOR MOUNTED | 399 | 387 | 10:1 | 97.0 | 110 | 140 | 33 | 8 | 4 | 4 | 120/1 | 3.0 | 3.8 | 400 | 1,2,3,4,5,6 |
| B-3 | MECH ROOM | CONDENSING FLOOR MOUNTED | 399 | 387 | 10:1 | 97.0 | 110 | 140 | 33 | 8 | 4 | 4 | 120/1 | 3.0 | 3.8 | 400 | 1,2,3,4,5,6,7 |
| | | | | | | | | | | | | | | | | | |

REMARKS:
1. PROVIDE NATURAL GAS, LOW NOx BOILER, LOW CO MODULATING BOILER.
2. HOT WATER TEMPERATURES: 110°F EWT/140°F LWT.
3. PROVIDE NEUTRALIZATION TANK FOR BOILER CONDENSATE, PIPED TO FLOOR DRAIN.
4. PROVIDE BOILER MANAGEMENT PANEL, TO CONTROL INTERNAL BOILER FUNCTIONS. PROVIDE BACNET MS/TP CARD FOR EXTERNAL BACNET CONNECTION, ALARM, ENABLE/DISABLE AND PUMP CONTROL.
5. PROVIDE WITH POLYPROPYLENE (PP) 4" INTAKE AND 4" EXHAUST KITS IN ACCORDANCE WITH MANUFACTUER'S RECOMMENDATIONS FOR ROOF INSTALLATION.
6. MANUFACTURER OPTIONS TO INCLUDE: FLOW SWITCH, CONSTANT SPEED PRIMARY PUMP.
7. PROVIDE ONE (1) SPARE PUMP.

## PUMP SCHEDULE

| MARK | SERVICE | TYPE | FLOW (GPM) | HEAD (FT. WG) | BHP | HP | VOLT/PH | RPM | REMARKS |
|---|---|---|---|---|---|---|---|---|---|
| PCHWP-1 | PRIMARY CHILLED WATER (CH-1) | INLINE | 213 | 30 | 2.2 | 3.0 | 460/3 | 1800 | 1,2,5 |
| PCHWP-2 | PRIMARY CHILLED WATER (CH-2) | INLINE | 213 | 30 | 2.2 | 3.0 | 460/3 | 1800 | 1,2,5 |
| PCHWP-3 | PRIMARY CHILLED WATER (STANDBY) | INLINE | 213 | 30 | 2.2 | 3.0 | 460/3 | 1800 | 1,2,5 |
| SCHWP-1 | SECONDARY CHILLED WATER | INLINE | 322 | 77 | 7.6 | 10.0 | 460/3 | 1800 | 1,2,5 |
| SCHWP-2 | SECONDARY CHILLED WATER (STANDBY) | INLINE | 322 | 77 | 7.6 | 10.0 | 460/3 | 1800 | 1,2,5 |
| SHWP-1 | SECONDARY HOT WATER SYSTEM | INLINE | 77 | 40 | 1.2 | 1.5 | 460/3 | 1800 | 1,2,5 |
| SHWP-2 | SECONDARY HOT WATER SYSTEM | INLINE | 77 | 40 | 1.2 | 1.5 | 460/3 | 1800 | 1,2,5 |
| PHWP-1 | PRIMARY HOT WATER SYSTEM (B-1) | INLINE | 33 | 20 | - | 0.5 | 277/1 | 1800 | 3,4 |
| PHWP-2 | PRIMARY HOT WATER SYSTEM (B-2) | INLINE | 33 | 20 | - | 0.5 | 277/1 | 1800 | 3,4 |
| PHWP-3 | PRIMARY HOT WATER SYSTEM (B-3) | INLINE | 33 | 20 | - | 0.5 | 277/1 | 1800 | 3,4 |
| | | | | | | | | | |

REMARKS:
1. PROVIDE WITH VFD AND BACNET MS/TP CARD.
2. PUMP IS SIZED FOR TOTAL SYSTEM CAPACITY WITH BACK-UP OR STANDBY PUMP FOR N+1 OPERATION, WITH LEAD/LAG CONTROLS.
3. PROVIDE WITH BACNET MS/TP CARD.
4. VERIFY FLOW AND HEAD REQUIREMENTS FOR SELECTED BOILER TO ENSURE PUMP IS SELECTED WITHIN 5% OF PEAK EFFICIENCY.
5. PROVIDE PUMP WITH SUCTION DIFFUSER AND FLEX CONNECTORS.

2

## CONTROL VALVE SCHEDULE

| EQUIPMENT | FLOW RATE (gpm) | NOMINAL SIZE (in) | MIN. CV | MAX. PD (psi) | SERVICE | CONFIGURATION | ACTION | EQUIPMENT | FLOW RATE (gpm) | NOMINAL SIZE (in) | MIN. CV | MAX. PD (psi) | SERVICE | CONFIGURATION | ACTION |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FCU-1-01 | 0.7 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-1-01 | 1.8 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-1-02 | 0.7 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-1-02 | 2.4 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-1-03 | 1.1 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-1-03 | 2.4 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-1-04 | 1.1 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-1-04 | 3.0 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-1-05 | 0.7 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-1-05 | 2.4 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-1-06 | 1.1 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-1-06 | 3.0 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-1-07 | 1.1 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-1-07 | 3.0 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-1-08 | 1.1 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-1-08 | 3.0 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-1-09 | 1.1 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-1-09 | 3.0 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-1-10 | 1.1 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-1-10 | 3.0 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-1-11 | 1.8 | 3/4 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-1-11 | 8.3 | 1 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-1-12 | 1.1 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-1-12 | 3.6 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-1-13 | 1.1 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-1-13 | 3.0 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-1-14 | 0.7 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-1-14 | 3.0 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-2-01 | 1.1 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-2-01 | 3.6 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-2-02 | 1.1 | 1/2 | 2.0 | 3.0 | HW | 3-WAY | MOD | FCU-2-02 | 3.6 | 3/4 | 2.0 | 3.0 | CHW | 3-WAY | MOD |
| FCU-2-03 | 1.1 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-2-03 | 3.0 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-2-04 | 1.1 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-2-04 | 3.0 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-2-05 | 1.1 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-2-05 | 3.0 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-2-06 | 1.1 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-2-06 | 3.0 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-2-07 | 1.1 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-2-07 | 3.0 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-2-08 | 1.1 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-2-08 | 3.0 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-2-09 | 0.7 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-2-09 | 1.8 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-2-10 | 1.1 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-2-10 | 3.0 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-2-11 | 1.4 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-2-11 | 4.8 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-2-12 | 0.7 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-2-12 | 2.4 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-3-01 | 1.1 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-3-01 | 4.2 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-3-02 | 1.1 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-3-02 | 3.6 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-3-03 | 1.4 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-3-03 | 4.2 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-3-04 | 1.1 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-3-04 | 3.6 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-3-05 | 1.4 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-3-05 | 4.2 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-3-06 | 1.1 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-3-06 | 3.6 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-3-07 | 1.1 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-3-07 | 3.6 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-3-08 | 1.4 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-3-08 | 4.8 | 3/4 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| FCU-3-09 | 1.4 | 1/2 | 2.0 | 3.0 | HW | 3-WAY | MOD | FCU-3-09 | 4.2 | 3/4 | 2.0 | 3.0 | CHW | 3-WAY | MOD |
| FCU-3-10 | 1.1 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | FCU-3-10 | 3.9 | 3/4 | 2.0 | 3.0 | CHW | 3-WAY | MOD |
| DOAS-1 PREHEAT | 13.3 | 1 1/4 | 2.0 | 3.0 | HW | 2-WAY | MOD | DOAS-1 | 74.8 | 3 | 2.0 | 3.0 | CHW | 2-WAY | MOD |
| DOAS-1 REHEAT | 6.2 | 1 | 2.0 | 3.0 | HW | 2-WAY | MOD | | | | | | | | |
| DOAS-2 PREHEAT | 10.6 | 3/4 | 2.0 | 3.0 | HW | 3-WAY | MOD | DOAS-2 | 59.5 | 3 | 2.0 | 3.0 | CHW | 3-WAY | MOD |
| DOAS-2 REHEAT | 4.9 | 3/4 | 2.0 | 3.0 | HW | 2-WAY | MOD | | | | | | | | |
| DOAS-3 PREHEAT | 11.6 | 1 1/4 | 2.0 | 3.0 | HW | 3-WAY | MOD | DOAS-3 | 65.3 | 3 | 2.0 | 3.0 | CHW | 3-WAY | MOD |
| DOAS-3 REHEAT | 5.4 | 3/4 | 2.0 | 3.0 | HW | 2-WAY | MOD | | | | | | | | |
| UH-1 | 1.2 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | | | | | | | | |
| UH-2 | 1.2 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | | | | | | | | |
| UH-3 | 1.2 | 1/2 | 2.0 | 3.0 | HW | 2-WAY | MOD | | | | | | | | |

M-603


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DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND

MARINE CORPS BASE

CAMP LEJEUNE, NORTH CAROLINA

MECHANICAL REPAIRS
BUILDING TT113

MECHANICAL SCHEDULES

DES. LWM
DR. PJR
CHK. JDL
SUBMITTED BY:
DESIGN DIR. JENNI REED
APPROVED: PWO OR OICC DATE
SATISFACTORY TO: DATE

SIZE E1 | CODE IDENT. NO. 80091 | NAVFAC DRAWING NO. 60042095
CONSTR. CONTR. NO. N40085-24-F-5940
SCALE AS NOTED | SPEC. | SHEET 88 OF 119

| REVISIONS | | | |
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| SYM. | DESCRIPTION | DATE | APP. |
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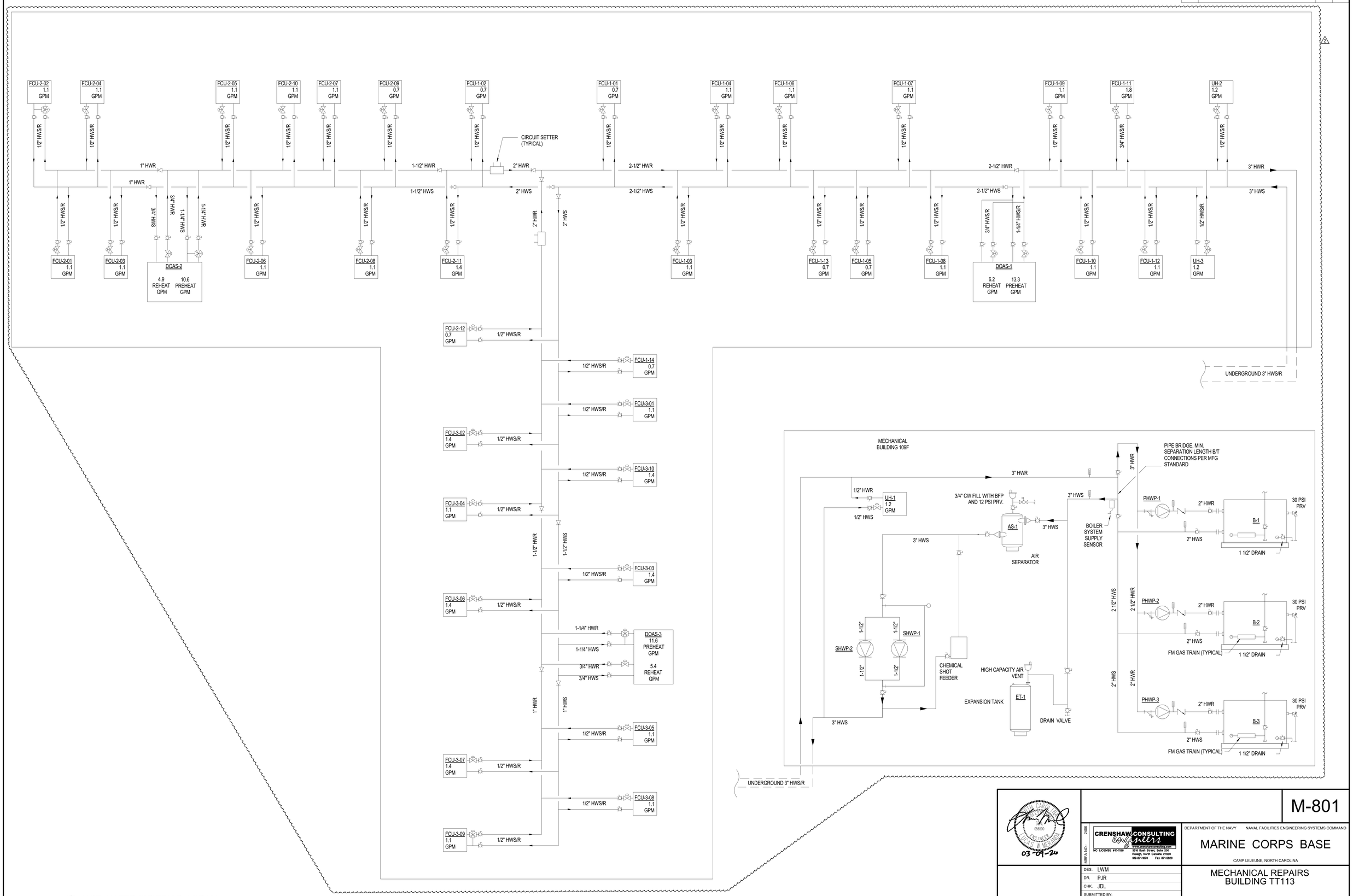


A1 HOT WATER SYSTEM DIAGRAM
NTS

M-801

03-09-26

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MARINE CORPS BASE

CAMP LEJEUNE, NORTH CAROLINA

MECHANICAL REPAIRS
BUILDING TT113

HOT WATER SYSTEM DIAGRAM

DES. LWM
DR. PJR
CHK. JDL
SUBMITTED BY:
DESIGN DIR. JENNI REED
APPROVED: PWO OR OICC DATE
SATISFACTORY TO: DATE

SIZE E1 | CODE IDENT. NO. 80091 | NAVFAC DRAWING NO. 60042102
CONSTR. CONTR. NO. N40085-24-F-5940
SCALE AS NOTED | SPEC. | SHEET 95 OF 119

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
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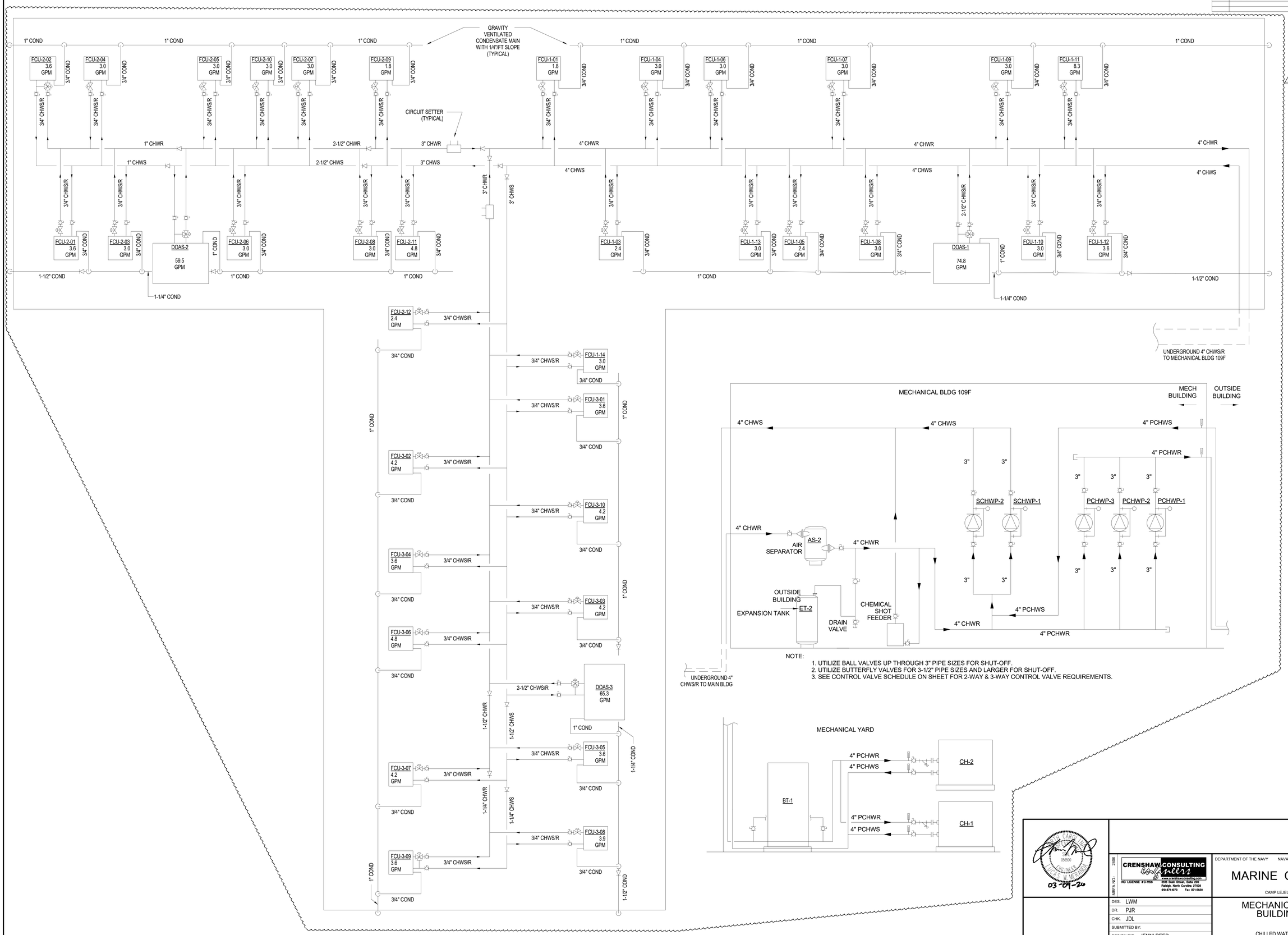


NOTE:

1. UTILIZE BALL VALVES UP THROUGH 3" PIPE SIZES FOR SHUT-OFF.
2. UTILIZE BUTTERFLY VALVES FOR 3-1/2" PIPE SIZES AND LARGER FOR SHUT-OFF.
3. SEE CONTROL VALVE SCHEDULE ON SHEET FOR 2-WAY & 3-WAY CONTROL VALVE REQUIREMENTS.

A1 CHILLED WATER SYSTEM DIAGRAM
NTS

03-09-26

M-802

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MARINE CORPS BASE

CAMP LEJEUNE, NORTH CAROLINA

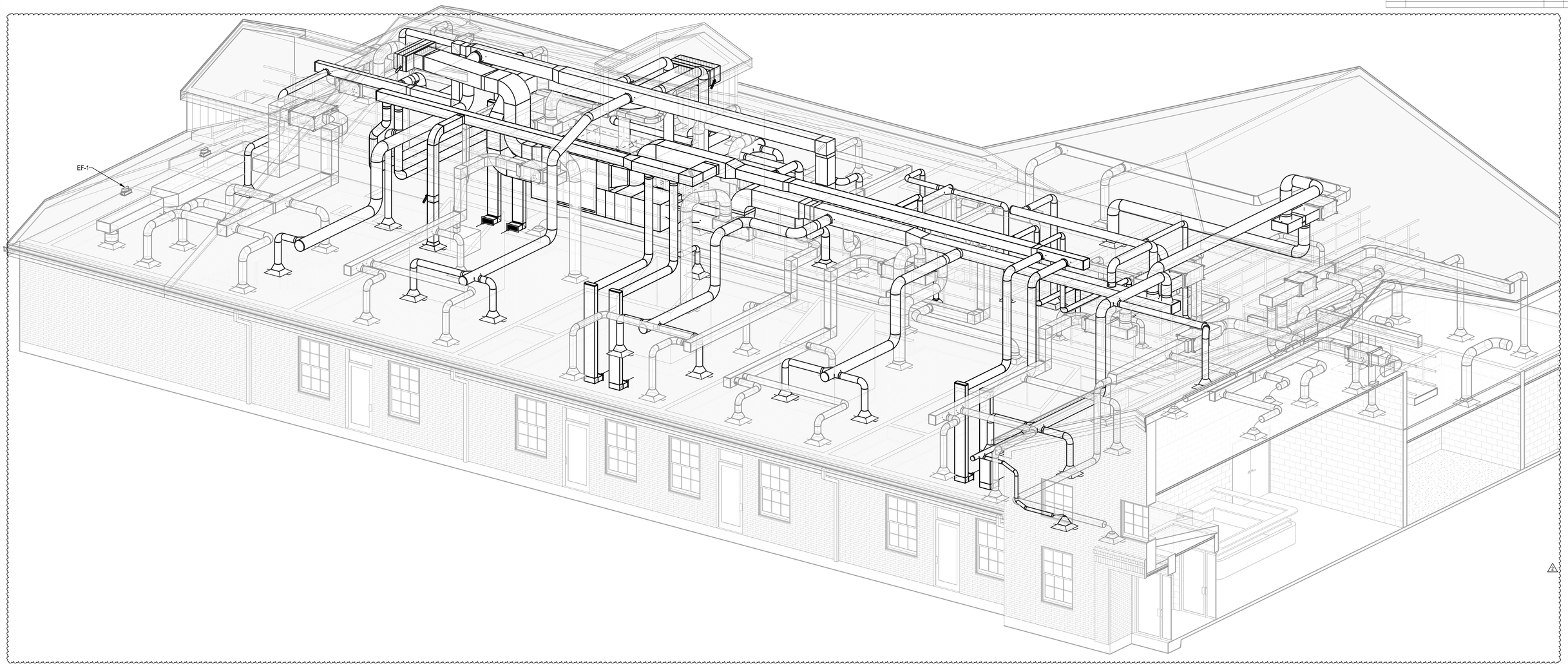
MECHANICAL REPAIRS BUILDING TT113

CHILLED WATER SYSTEM DIAGRAM

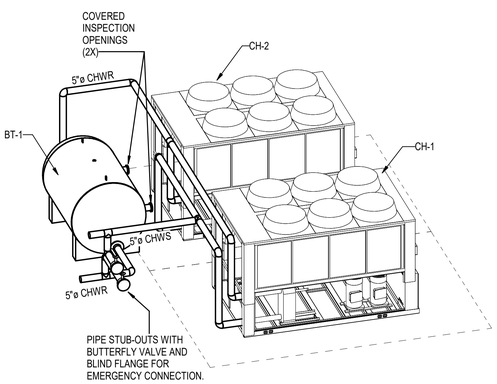
DES. LWM
DR. PJR
CHK. JDL
SUBMITTED BY:
DESIGN DIR. JENNI REED
APPROVED: PWO OR OICC — DATE
SATISFACTORY TO: — DATE

SIZE E1 | CODE IDENT. NO. 80091 | NAVFAC DRAWING NO. 60042103
CONSTR. CONTR. NO. N40085-24-F-5940
SCALE AS NOTED | SPEC. | SHEET 96 OF 119

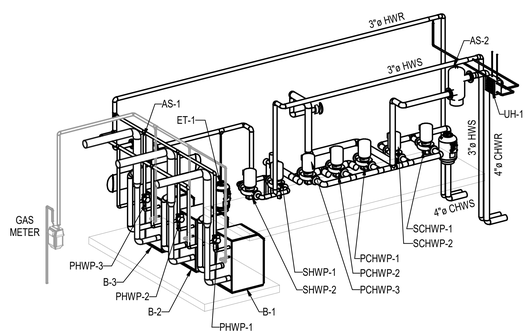
| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| 2 | PPI REVISION 2 | 03.09.26 | |




B1 3D ISOMETRIC VIEW - DOAS-1




A1 3D ISOMETRIC VIEW - CHILLER YARD




A3 3D ISOMETRIC VIEW - MECHANICAL BUILDING

M-901

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CAMP LEJEUNE, NORTH CAROLINA

03-01-26

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DR. PJR
CHK. JDL
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MECHANICAL REPAIRS
BUILDING TT113

3D ISOMETRIC VIEW - DOAS-1

SIZE E1 | CODE IDENT. NO. 80091 | NAVFAC DRAWING NO. 60042104
CONSTR. CONTR. NO. N40085-24-F-5940
SCALE AS NOTED | SPEC. | SHEET 97 OF 119

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| 2 | PPI REVISION 2 | 03.09.26 | |
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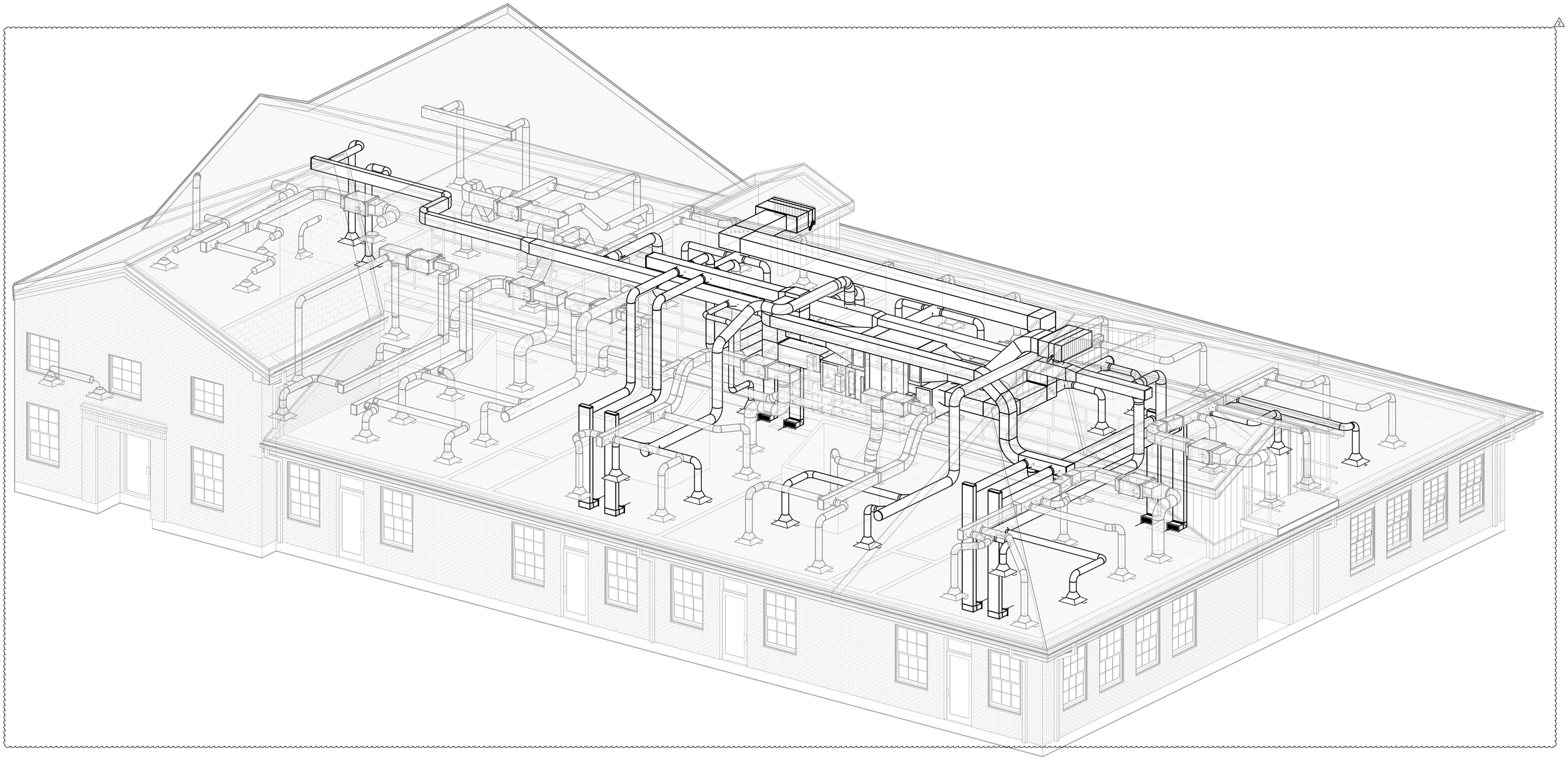

A1 3D ISOMETRIC VIEW - DOAS-2

M-902

03-09-26

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DESIGN DIR. JENNI REED
APPROVED: PWO OR OICC DATE
SATISFACTORY TO: DATE

MECHANICAL REPAIRS
BUILDING TT113

3D ISOMETRIC VIEW - DOAS-2

| SIZE | CODE IDENT. NO. | NAVFAC DRAWING NO. |
|---|---|---|
| E1 | 80091 | 60042105 |
| | | CONSTR. CONTR. NO. N40085-24-F-5940 |

SCALE AS NOTED SPEC. SHEET 98 OF 119

| REVISIONS | | | |
|---|---|---|---|
| SYM. | DESCRIPTION | DATE | APP. |
| 2 | PPI REVISION 2 | 03.09.26 | |
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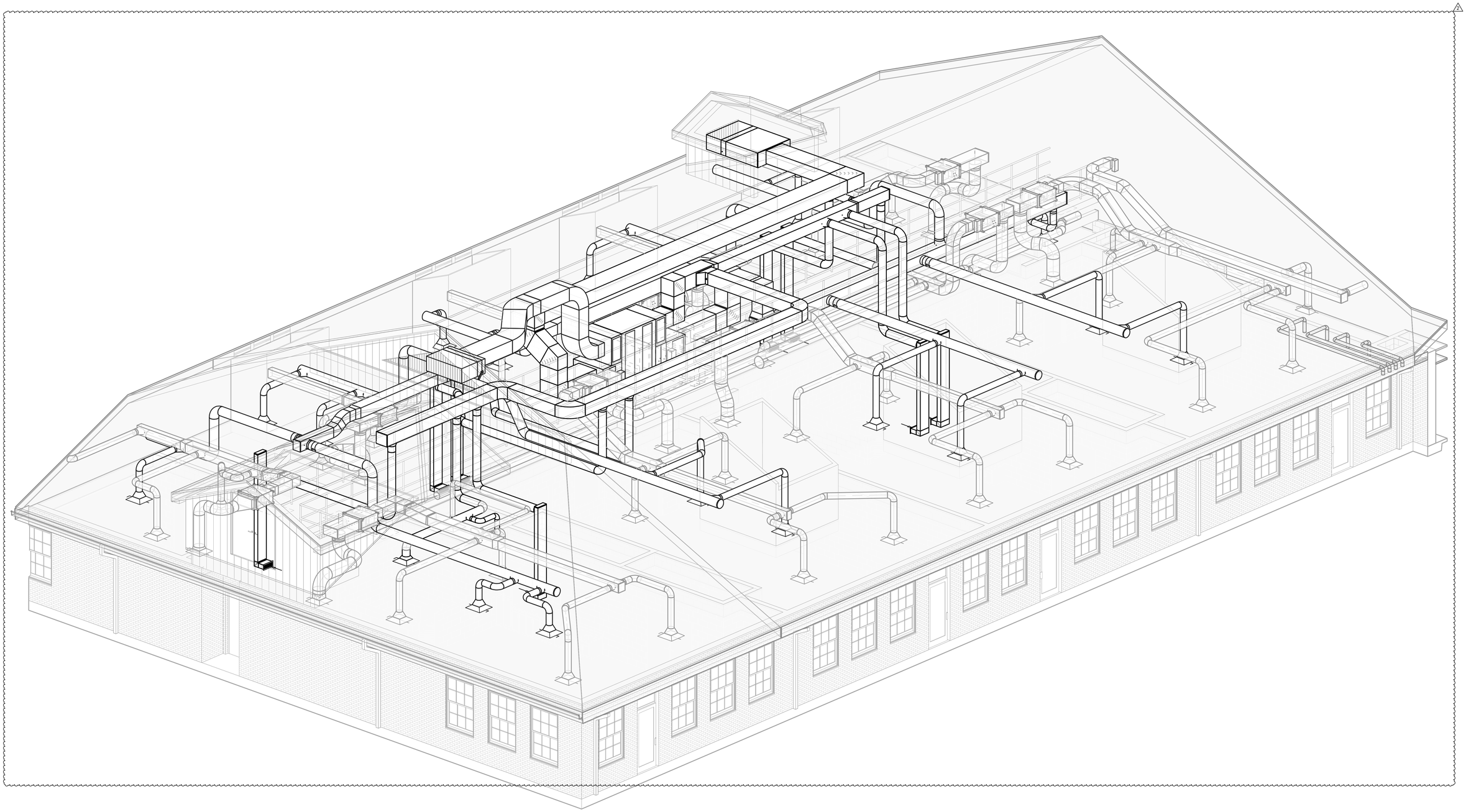

A1 3D ISOMETRIC VIEW - DOAS-3

M-903

03-09-26

MBFA NO: 2406

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3D ISOMETRIC VIEW - DOAS-3

| SIZE | CODE IDENT. NO. | NAVFAC DRAWING NO. |
|---|---|---|
| E1 | 80091 | 60042106 |
| | | CONSTR. CONTR. NO. N40085-24-F-5940 |
| SCALE AS NOTED | SPEC. | SHEET 99 OF 119 |